1. Excellence

1.1 Objectives

PHOTONFOOD's aim is to provide a portable solution for flexible farm-to-fork sensing of microbial and chemical contamination in food products and along the food production chain. PHOTONFOOD will develop an integrated solution that combines innovations in mid-infrared (MIR) sensing with smart paper-based sample treatment, and advanced data analysis. This solution will be developed for the detection of microbial and chemical contamination, validated and demonstrated in real scenarios along the food value chain (Fig. 1). Novel light emitting diode (LED) light sources, as well as interband and quantum cascade lasers (ICL/QCL), will be combined with sophisticated waveguide technology to transform MIR sensing from existing lab solutions into a portable solution. The combination of MIR sensing with a paper microfluidic sample handling device is especially powerful, as it provides an easy and affordable means to measure purified extracts of contaminants. Purified extracts are necessary if one wants to achieve quantitative spectroscopic determination of levels of chemical and microbial contamination in food matrices at relevant detection limits.

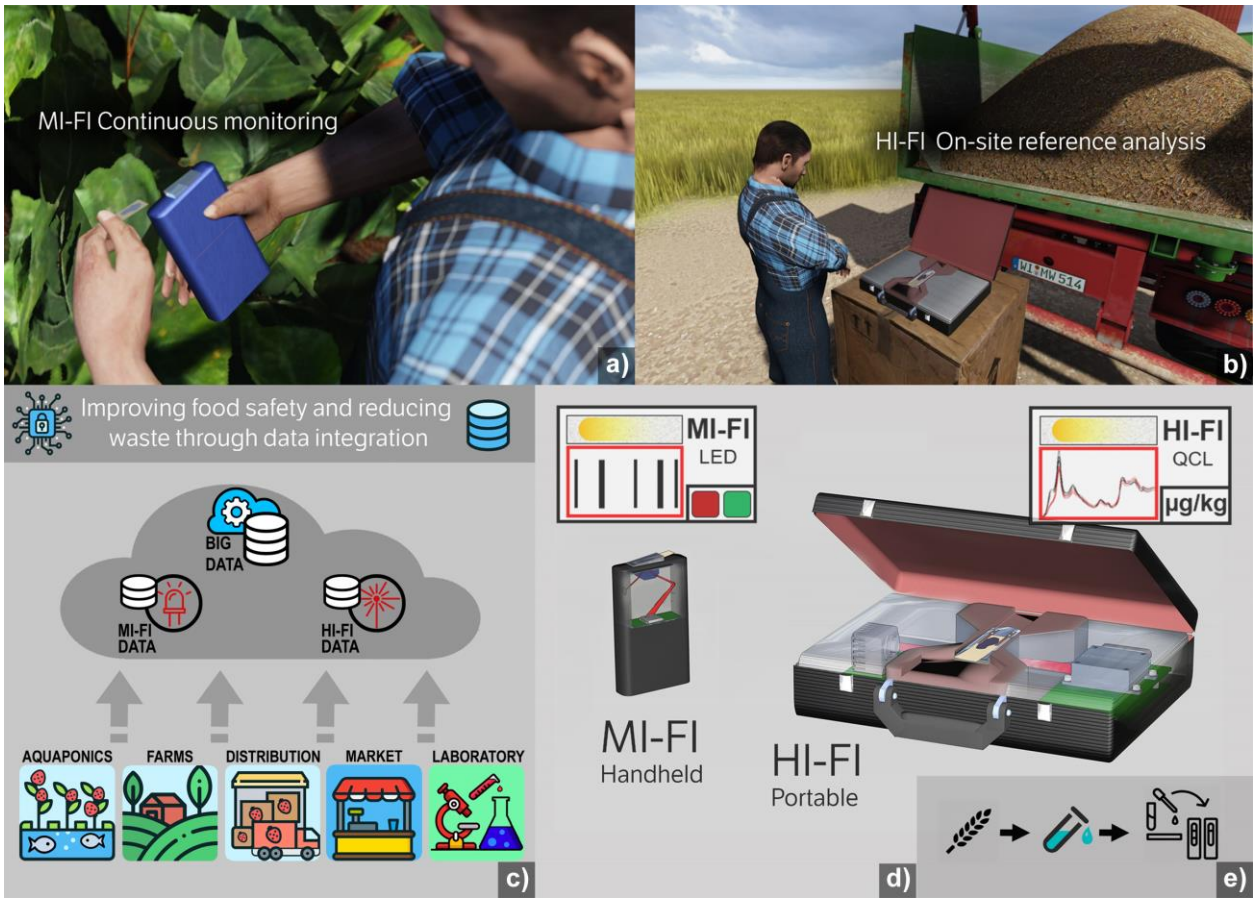

Figure 1: Overview of the PHOTONFOOD project. (a) The use of the MI-FI device for daily monitoring, and (b) the HI-FI device for reference analysis along the value chain. Compared to existing testing off-site, considerably more data will be collected. (c) The transfer of the data to platforms such as Food Trust™ and data integration will define a new paradigm for food safety control and reduction of food waste, and thereby represent a disruptive technology in agronomy and food industries. (d) PHOTONFOOD devices will feature two different types of photonic technologies (IC-LED and ICL/QCL), and (e) a paper-based microfluidic sample handling unit, for detection of minute concentrations of contaminants.

PHOTONFOOD will develop and demonstrate a photonic based monitoring solution for highly important contamination problems in the food production chain, namely **(i) fungal and water mould (oomycetes) contamination and mycotoxins in wheat, nuts, dried fruits and aquaponic-based herbs, and (ii) pesticides and antibiotics in aquaponic-based herbs**. Hence, the applicability of the developed portable devices will be demonstrated on-site at the end user level (farms and retail), including aquaponics as a novel type of food production system. We will demonstrate that the PHOTONFOOD solution is highly flexible and can be easily diversified: it can be used at the most critical control points in the production and distribution of these products,

such as the entry point of the European Union for products produced outside the EU, at the farm level (traditional farming, novel forms of farming and on-farm vending), at the distributer level, and the level of the retailer.

The solution is based on **two innovative photonic devices**, that are suitable for a wide range of application scenarios. Both photonic devices will have integrated **low-cost paper-based microfluidics** for sample collection, pretreatment, and analyte preconcentration, as well as an integrated **data analysis platform** (see Fig. 1). The first photonic device will be **low-cost and handheld**, delivering medium-fidelity data (**MI-FI device** see Fig. 1) by incorporation of innovative MIR light emitting diode (LED) light sources based on interband cascade (IC) technology. The second photonic device will be **medium-cost and portable**, delivering high-fidelity data (**HI-FI** device, see Fig. 1) by incorporation of interband and quantum cascade lasers (ICL/QCL). The **HI-FI device** will provide high-fidelity molecular signatures that can be used for **reference measurements**. The **handheld MI-FI device** is tailored for daily **monitoring** along the whole food chain and even retail, which is feasible due to its low envisioned price. The HI-FI spectra will allow the quantification of contaminations at relevant detection limits and will provide sufficient precision for accreditation of the method. Advanced data analysis will be used to calibrate the MI-FI and HI-FI spectral data. The obtained data will be further utilized by transferring it to and integrating it into platforms such as the block chain-based IBM Food Trust™ data platform for tracing food contamination and improving food quality. The availability of big data on food safety and the integration of the data into food block chain platforms will define a new paradigm for food safety control and waste reduction (see Fig. 1).

The importance of detecting fungal and mycotoxin contamination. Fungi are among the most resilient spoilage microorganisms and can be missed by the most advanced control safety measures utilized in food industry[1]. Fungi are rapidly dispersed by water and air, can survive under extremely demanding conditions such as thermal, detergent- and preservatives-based processing regimes used in modern farm-to-fork production[2]. Approximately 25% of the global food supply is wasted or lost because of postharvest fungal spoilage[1]. In a survey of food manufacturers in the United States and Europe, 94% of the surveyed producers reported fungal spoilage in the finished product as a frequent problem[1]. The majority of cases were related to nuts (70%), followed by fruits and vegetables (12%), herbs and spices (8%) and cereals (4%). The main food spoilage fungi secrete toxic secondary metabolites, so-called mycotoxins of which the immunosuppressive deoxynivalenol and the carcinogenic aflatoxins are among the most prevalent[1]. The European Commission pointed out that 5–10% of global crops produced are lost annually due to mycotoxin contamination. Considering that the EU produces about 133 million tons of wheat annually, about €1.2-2.4 billion can be lost for wheat alone[3]. Mycotoxins are the most frequently notified food safety hazard in the EU[1]. During the last five years, the number of mycotoxin notifications has been increasing at an average rate of 15% per year, thus highlighting the severity of the problem[1]. During this five-year period, the EU´s Rapid Alert System of Food and Feed (RASFF) reported more than 300 alerts relating to aflatoxin levels exceeding legal limits in peanuts imported into the EU[3]. Moreover, fungal and fungal-like oomycetes (water mould) contaminations constitute an increasingly urgent problem in aquaponics and vertical farming, a rapidly developing novel plant and vegetable production systems[4]. Characteristic to aquaponic and hydroponic plant culture is the continuous presence of water in the production system, where the humid or aquatic environment suits most fungal and oomycetes plant pathogens[4]. Strikingly, our partner, the aquaponic company BIGH reported direct losses of aromatic herbs of around 5000 pots. BIGH estimates that it could increase the production by additional 20 000 pots if they were not forced to reduce their production density due to contamination by oomycetes (downy mildew). Currently, mycotoxins and fungal detection is addressed by laborious reference analytical methods, whereby for fungal detection methods based on culturing with subsequent genetic analysis are applied. Mycotoxins are usually detected via sophisticated lab-based LC-MS/MS-based multi-metabolite methods. Immunoassays (ELISA) are also applied for the rapid determination of a few regulated mycotoxins, especially in industrial settings. However, ELISA protocols still require several sample preparation steps, are not flexible, can hardly be used on-site, and are still prone to matrix effects due to cross reactivities of the employed antibodies. Among of the more rapid methods, lateral flow strips coul be used even on sites but their price is ill far away from being optimal. The relevant limit of detection (LOD) for mycotoxins will vary depending on

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the type of mycotoxin, for example the case of deoxynivalenol (DON) the LOD is around 1000 ppb. Therefore, prevention and control of these toxic secondary metabolites remains difficult, and the agriculture and food industries continue to be vulnerable to problems of contamination.

The importance of detecting pesticide residuals in cereals, vegetables and fruits. Pesticides are used in agriculture to satisfy a continuously increasing consumer demand for agricultural products by minimizing the pre- and post-harvest losses. While agriculture productivity is increasing, pesticides represent hazards to consumers since their residues accumulate in different fractions in agriculture products such as cereals, vegetables and fruits. The persistence of pesticides has been reported for wheat products, where milling distributes pesticide residues in different fractions in the bran, germ, and endosperm. In addition, pesticide residues could be detected in flower and in bread[5]. In general, food processing allows decrease in pesticide levels. However, in some cases, residue levels of pesticides even increased in the final product due the processing of the raw materials[6]. This can be related to water removal, for example in the production of dry fruits such as raisins, prunes and apricots[6]. The control of pesticides residues in such novel productions as aquaponics has a particular importance, since pesticides and herbicides are toxic for fish[7].

Today, the detection of pesticides residues is time-consuming and performed by the traditional instrumental chemistry techniques, such as gas chromatography (GC), liquid chromatography (LC) or capillary electrophoresis (CE), coupled to different detectors, especially to mass spectrometers (MS). While several biosensors were developed for the quick detection of pesticides, none of them have been accredited for the lab use and they are not broadly available on the market. The main reason for that is that relevant detection limit for pesticides is in 0.1 µkg/g which is not achievable by novel biosensors.

The importance of detecting antibiotics in aquaponics. Aquaponics integrates recirculating aquaculture producing fish and hydroponics producing plants. In standard aquaculture, antibiotics are regularly used for fish production in order to eliminate the risk of fish disease. However, antibiotics are not allowed in aquaponics because they may impact plant growth, as well as quality and the maintenance of nutrient solubilizing microorganisms. Therefore, the control of the presence of antibiotics in water circulating in aquaponics farms is of significant importance.

Currently, detection of antibiotics residues is performed by the EU four-plate (EU4PT) in an ampoule or tube test or the Premi®Test (DSM) method which is based on microbiological assessment[8]. Both tests are time consuming. During the last decade, a series of high-throughput rapid methods for detecting antibiotics residues in food were developed, most of which are based on immunoassays, microbiological tests and biosensors. Despite their throughput, the detection limit is still above the requirements[9].

Up to date, only few photonic techniques have been adopted for the routine detection of contaminations such as fungi, mycotoxins, pesticides and antibiotics[10,11,12], despite their potential for detecting microbial and chemical contaminants, their modularity, and their easy adaptability to a wide range of deployment scenarios. Among the optical/spectroscopic methods, **MIR spectroscopy** has proven to be by far the most reliable analytical technique for probing molecular fingerprints of microbial and chemical contaminations via molecular spectral signatures[13,14]. The **Norwegian University of Life Sciences (NMBU)** has developed a fully automated laboratory system for spectroscopy-based analysis of fungi. The research is based on the R4SME FP7 project FUST, which was coordinated by the PI (see Table 4). **Ulm University (UULM), BOKU and IRIS** have developed a MIR sensing platform for lab-based analysis of mycotoxins in the MYCOSPEC FP7 project coordinated by IRIS (see Table 4). *Both projects demonstrated that laboratory-based MIR spectroscopy can be used for monitoring fungi and mycotoxins in different food processing environments.* However, portability and analysis of trace concentrations of contaminations in food matrixes remains a challenge to be solved. While few studies report the detection of food contaminants at low concentrations by indirect detection (detection of food matrix changes), a reliable detection of contamination at low

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concentrations (e.g. at ppm level) is theoretically not possible. Slightly varying food matrices introduce molecular signatures that completely obscure the levels of contaminations that need to be detected and that are at the parts per million level. Therefore, a pre-concentration of the contaminants is necessary in order to achieve the detection levels needed for most of the contaminants, and, eventually, to develop a portable photonic solution for detecting food contaminants on farms and in food production lines, wholesale, and retail. **Wageningen University** is currently developing user-friendly and inexpensive devices based on 3D printing and paper microfluidics in the EU ETN project FoodSmartphone (see Table 4). Such solutions can be further developed for pre-concentration of tiny amounts of contaminants in food matrices.

PHOTONFOOD will build on the development of photonic components, modules and subsystems pioneered in FUST, MYCOSPEC and MIRACLE (see Table 4), and user-friendly sample handling from FoodSmartphone. PHOTONFOOD will develop a flexible portable solution for the detection of microbial and chemical contamination. The solution will include high-fidelity (HI-FI) and mid-fidelity (MI-FI) MIR photonic devices, coupled to paper-based sample preparation module, and will be benchmarked against state-of-the-art laboratory platforms. Advanced data analysis will be used for calibrating the spectral MI-FI and HI-FI data and for exploiting the obtained data for food safety and waste reduction. The developed PHOTONFOOD photonic solutions are broadly adaptable for usage along the entire food production value chain, from the farmer and distributer, to the importer, and up to the supermarket and consumer level.

The research will be performed by a multidisciplinary team with globally acknowledged leading expertise in the field of IR spectroscopy, light source and waveguide development, paper-based on-site chemical analysis, food spoilage microbiology, and data science. The project will involve collaboration with leading mycotoxin analysis laboratories, an aquaponics farm, a distributor of dry foods, farms with on-site food processing and vending, and a producer of a digital platform for food safety.

The PHOTONFOOD solution will be validated and demonstrated in real settings for the detection of fungal and chemical contaminations by involving relevant major stakeholders in the food value chain, an international analytical laboratory specialized on mycotoxins and the Hungarian Food Safety Laboratory, and achieve a **Technology Readiness Level (TRL) of 6.** This will allow the developed photonic solution to have a short and direct way into the market after the project completion. The developed solution will in future allow the detection of contamination points in the food value chain and reduce food production losses and food waste. The development of core photonics and microfluidic technologies in the PHOTONFOOD project will in addition facilitate innovation in other application areas and – next to food safety - harness Europe at the forefront of photonics innovation and sensing/monitoring technology.

Specific objectives and related key success criteria

The main objective of the PHOTONFOOD project is to develop and demonstrate in real settings a flexible MIR photonic solution for the detection of microbial and chemical contaminations based on **innovative photonic devices, paper-based sample handling and advanced data mining solutions.**

The transfer of the same core technology (i.e., MIR photonics) from the laboratory bench to a flexible portable solution enables broad usage on farms, in food processing, wholesale and retail with the unique advantage of providing the same analytical signatures, at two different levels of 'fidelity', (i) a HI-FI device that can be used for on-site reference analysis and (ii) a MI-FI device that can be used for contamination monitoring. This implies that several technical challenges involving **efficient microfluidic sampling, extraction and preconcentration combined with advanced MIR readout technologies, tailored light sources and waveguides, adaptable photonic devices, and data processing/mining tools for the detection of fungal/mycotoxin and chemical contaminations must be overcome.** The current technical challenges will be addressed as follows:

Table 1: PHOTONFOOD specific technical challenges, specific objectives (SO) and key success criteria.

Paper-based microfluidic preconcentration and separation
Challenge: Due to the detection limits defined by the EU regulations, the PHOTONFOOD solutions would need to have a detection limit for microbial and chemical contaminants at EU regulation levels and it would be essential to obtain purified and enriched solutions of contaminants for spectroscopic detection. **PHOTONFOOD will deliver smart, cheap and user-friendly paper microfluidic devices to extract, separate and preconcentrate chemical and microbial contaminants prior to spectroscopic analysis.** This solution can be adapted to allow for trace analysis of any food contaminant.

SO1: To develop paper microfluidic sampling, extraction and pretreatment for food contaminants from hard and soft commodities (WP2, WP6).
Key success criteria: • Sampling strategies defined for hard and soft samples and delivered as a report (D2.1 M24). • Purification and enrichment device for one target mycotoxin (DON) in one commodity (wheat) demonstrated in real settings (D2.1 M24). • Diversification of approach to at least a second target contaminant in a second commodity demonstrated (D2.1 M48).
3D printed interface for microfluidics and optical surfaces
Challenge: In order to optimize the measured signal by the portable MIR sensing solution, the contact between the separated analytes and the MIR sampling surfaces and the detection area of the paper-based microfluids needs to be optimized. **3D-printed cartridges will be iteratively designed and produced to ensure alignment in the interface between the paper microfluidic strip and the optics for readout**. The technology will be universally applicable for any chemical signal enhancement in optical measurements in medical and life sciences.
SO2: To develop a functionalized cartridge for the paper microfluidic devices that allows robust and user-friendly interfacing with spectroscopic detection (WP1, WP2).
Key success criteria: • Report on strategy for coupling optical components, and the paper microfluidic interface (D2.1 M24). • Fully integrated device for analysis of one target mycotoxin (DON) in one commodity (wheat) available (D2.2 M48). • Diversification of approach to at least a second target contaminant in a second commodity finalized (D2.3 M48)..
HI-FI ICL/QCL light source for 1500-1800 cm^{-1} range spectral range
Challenge: In order to develop a high-fidelity (HI-FI) portable MIR spectroscopic device providing the level of sensitivity and selectivity needed for the detection of relevant fungal and mycotoxin contamination at defined detection limits in food/commodity matrices, the brightest possible MIR light sources are required covering a broad spectral window. PHOTONFOOD will develop **monolithic interband/quantum cascade laser (ICL/QCL) technologies with the required coverage of a broad wavelength window.** A **monolithic QCL or ICL** is one of the major photonics innovations of PHOTONFOOD and will be useful for a plethora of applications within the field of photonics.
SO3: To develop tunable ICL/QCL MIR lasers for measurements with high precision and sensitivity (WP3).
Key success criteria: • Broadly tunable lasers based on Vernier effect demonstrated for the 1500-1800 cm^{-1} wavelength range (D3.2 M26). • Tunable ICLs/QCLs available and validated in operation within the HI-FI device (D4.3 M24).
MI-FI IC-LED light sources for 1500-1800 cm^{-1} range spectral region
Challenge: For low-cost handheld mid-fidelity (MI-FI) analyzers, a low-cost powerful light source in the MIR needs to be developed. PHOTONFOOD will develop **MIR light emitting diodes (MIR-LEDs)** with significantly lower production cost with lesser spectral brightness and a broader emission linewidth as trade-off. The availability of MIR IC-LED technology across the MIR spectral window would provide a breakthrough alternative to single mode laser technology facilitating low-cost hand-held analyzer platforms with numerous applications across a wide range of monitoring scenarios.
SO4: To develop IC-LED light sources for low-cost contamination detection (WP3).
Key success criteria: • IC-LEDs (Interband cascade light emitting diodes) for integration into MI-FI device available (D3.1 M18). • IC-LEDs available and validated in operation within MI-FI devices (D4.2 M30).
HI-FI MIR portable photonic device in the 1500-1800 cm-1 range
Challenge: For the on-site detection and reference analysis of the level of microbiological and chemical contaminants in food commodities, portable label-free MIR photonics read-out devices are needed. PHOTONFOOD will develop a **high-end portable ICL/QCL laser-based device (HI-FI)** based on thin-film nanocrystalline diamond waveguides to provide MIR molecular signatures of commodity matrix changes and contaminants with maximum spectral resolution, sensitivity and precision at a **reference level**.

In addition to the food sector, the portable devices will be applicable for a broad range of applications, such as wastewater or liquid fertilizer analysis.
SO5: To prototype a HI-FI MIR photonic sensing device and integrate ICL/QCL light sources (WP3).
Key success criteria: • Thin-film GaAs waveguide- based photonic sensing platform demonstrated (D4.1 M18). • Robust sensing interface using on-chip nanocrystalline diamond thin-film waveguide technology established (D4.1 M18). • One portable HI-FI ICL/QCL analyzer platform (1500-1800 cm^{-1}) validated, tested and demonstrated (D4.3 M24).
MI-FI MIR handheld photonic device for the 1500-1800 cm^{-1} range
Challenge: For daily monitoring of contamination a low-cost handheld device is needed. PHOTONFOOD will develop **a low-cost monolithic dispersive MIR device (MI-FI) using MIR IC-LEDs** obtaining rapid medium-resolution MIR molecular signatures with satisfactory precision and sensitivity. The handheld device will be applicable to a wide range of applications outside the food sector, such as monitoring of wastewater, process fluids, as well as surface and ground water.
SO6: To prototype MI-FI MIR photonic sensing devices and integrate MIR IC-LED light sources (WP1, WP4, WP6).
Key success criteria: • MIR IC-LEDs (1500-1800 cm^{-1} range) tested for MIR sensing applications (D4.3 M30). • Hand-held MIR IC-LED (1500-1800 cm^{-1} range) monolithic grating spectrometer prototype assembled and lab tested (D4.3 M30). • Five MI-FI MIR IC-LED (1500-1800 cm^{-1}) hand-held photonic sensing devices validated and demonstrated (D4.3 M30).
Advanced data analysis platform
Challenge: The use of a narrow spectral window (1500-1800 cm^{-1} range) provides a substantial challenge for the data processing. Further, the prediction of multiple contaminants obtained from multiple commodities requires stable calibration procedures. **PHOTONFOOD will innovatively resolve this by developing specific pre-processing tools and establishing robust calibration models for MI-FI and HI-FI data covering a narrow but very discriminative spectral range**. The developed approaches will serve as a platform for future photonic sensor applications based on IC-LEDs and ICL/QCL data.
SO7: To use advanced data analysis to calibrate MI-FI and HI-FI solution for detection of microbial and chemical contaminants in food (WP5, WP6).
Key success criteria: • Pre-processing approaches for spectroscopy data of MIR IC-LED and ICL/QCL light sources established (D5.1 M36). • Machine-learning based calibration and classification tools established (D5.2 M36). • Calibration and classification tools validated and demonstrated (D5.2 M36).
PHOTONFOOD software application for data analysis for mobile and laptop
Challenge: The toolbox of data processing algorithms needs to be accessible through a PC and a mobile application. The data analysis models will be **established in a cloud service and provided as application for a laptop and a mobile phone for the MI-FI and HI-FI instrumentation.** This will allow the demonstration of the broad applicability of the solution at the end of the project and enable fast commercialization of the solution after the project period. During the project period, PHOTONFOOD will develop a strategy for data integration into the IBM's Food Trust™ block chain. This will in future allow rapid mitigation of contamination events and is expected to reduce associated food waste considerably.
SO8: To provide user-friendly interface, intuitive decision support tools and data integration strategy (WP1, WP5).
Key success criteria: • Collection of user requirements summarized in a report (D1.1 M24). • Architecture design validation fulfilled (D5.4 M36). • Software platform demonstration and end-user acceptance (D5.5 M40).
Validation and demonstration
SO9: To validate and demonstrate the PHOTONFOOD solutions in real settings (WP1, WP6).
Key success criteria:

• PHOTONFOOD solutions demonstrated for fungal contamination and mycotoxins in wheat, nuts, dry fruits and aquaponic-based herbs (D1.2 M40, D6.3 M46, D6.4 M46). • The applicability of the PHOTONFOOD solution to pesticides and antibiotics in aquaponic-based herbs demonstrated (D1.2 M40, D6.3 M46, D6.4 M46).
Dissemination of the PHOTONFOOD solution to stakeholders
SO10: To assess marketability for the PHOTONFOOD solution. (WP1, WP7)
Key success criteria: • Cost assessment in line with target levels (D1.3 M36). • Positive initial feedback from end-user groups obtained (D1.4 M48). • Positive attitude among key stakeholder groups assessed in stakeholder survey (D1.4 M48). • TRL 6 reached. The device is accepted by end-users and key corporate decision-makers in immediately relevant market segments (D1.4 M48).
Exploitation of the application of the developed components in other sectors
SO11: To strengthen Europe's leadership in photonics (WP1).
Key success criteria: • Positive assessment of the applicability of the technology to other value chains beyond those in which the device is being validated (D1.4 M48). • Positive attitude among key stakeholder groups assessed in stakeholder survey (D1.1 M24).

1.2 Relation to the work programme

The project addresses topic ICT-37-2020 Advancing photonics technologies and application driven photonics components and the innovation ecosystem: Sub-topic Flexible Farm-to-Fork Sensing. Table 2 outlines how PHOTONFOOD fits to the specific topic challenge and scope, and to other relevant parts of the "Towards 2020: Photonics Driving Economic Growth in Europe – Multiannual Strategic Roadmap 2014-2020".

Table 2: Contribution of PHOTONFOOD to the ICT-37-2020 call and the Photonic Strategy Roadmap

Challenge I as defined in ICT-37-2020 call
"*To develop and apply core photonics technology for the next generation of devices (including components, modules and sub-systems) in order to drive innovation in key application areas, which are significant for current or future markets and where photonics can bring a key competitive advantage*."
PHOTONFOOD contribution
PHOTONFOOD will design, develop and validate next-generation MIR photonic sensing devices, based on innovative MIR light emitting diode (IC-LED) light sources and ICL/QCL lasers combined with novel waveguides, for the detection of microbial and chemical contamination in food and commodities along the food value chain.
Challenge II as defined in ICT-37-2020 call
"*Innovative photonic sensing solutions can contribute to reducing food production losses and food wastes, estimated in Europe at about 300 kg per capita, and to increasing food safety for the end consumer along the food production chain from farm to fork."*
PHOTONFOOD contribution
PHOTONFOOD will provide more and better data on contamination along the food production chain, since on-site monitoring data from the MI-FI device and reference data from the HI-FI device will be connected to existing data platforms such as the Food Trust™. The use of the PHOTONFOOD data about microbial and chemical contamination will in future allow detecting the weak points of the food chain. Thus, it will make a considerable contribution to the reduction of food wastes, where the reduction of wastes in cereals production and import of nuts and dry fruits will be about 40% and 60% in aquaponic herbs production of the existing losses associated with the fungal and mycotoxin contaminations.
Scope I as defined in ICT-37-2020 call
"Development of an innovative smart photonic sensor solution, utilizing an appropriate bandwidth between the ultraviolet (UV) and the far-infrared spectral range for monitoring food quality with respect to microbiological and chemical contamination along the farm-to-fork food production chain."
PHOTONFOOD contribution
PHOTONFOOD will provide the first low-cost hand-held MIR sensing solution (MI-FI) and a high-end portable MIR device (HI-FI) enabling flexible microbial and chemical analysis of food products on farms,

in food processing, wholesale and retail. The MIR region was chosen as the appropriate band widths since it has proven to be the most discriminative region for determination of chemical and microbial contaminants. The photonics sensing solution will contain an innovative smart paper-based microfluidic sampling pre-concentration allowing the detection of tiny amounts of contaminants.
Scope II as defined in ICT-37-2020 call
"The targeted solution should combine photonic sensing technology with advanced data analysis techniques and be portable, easy-to-use, flexible, and broadly adaptable for usage on farms, in food processing, wholesale and retail."
PHOTONFOOD contribution
PHOTONFOOD will innovatively combine the developed photonic sensing devices with advanced data analysis approaches for mining specific fungal and mycotoxin contaminants in the pre-concentrated and analyte-separated samples. PHOTONFOOD will develop robust calibration modelling strategies for the data obtained by the proposed photonic solutions using the most advanced pre-processing, machine learning and deep learning tools. The calibration models for the cases demonstrated in PHOTONFOOD will serve as a template for further developments beyond the project period as they will set standards in the data pre-processing and methodology to detect and utilize specific molecular signatures. The models will be established in a cloud service and provided as application for a laptop (HI-FI device), and as a mobile app (MI-FI device). PHOTONFOOD will further develop a strategy for data integration into existing platforms such as the Food Trust™ Block Chain. The developed technology will be validated and demonstrated for the usage on farms, food processing, wholesale and retail through involvement of the relevant partners.
Scope III as defined in ICT-37-2020 call
"Actions should focus on the following areas: (1) food production by small/medium-sized farms; (2) novel types of food production, such as aquaponics; (3) on-site food processing and vending (e.g. on farms or local food markets)."
PHOTONFOOD contribution
PHOTONFOOD focuses on (1) small/medium-sized cereal farms through NEBIH, nuts and dried fruit production farms through the distributor Seeberger; (2) novel type aquaponic and on-site food producer BIGH. PHOTONFOOD will perform extensive demonstration of HI-FI and MI-FI devices by Seeberger, end-user ROMER, food-safety authority NEBIH and aquaponics company BIGH. As a food safety authority, NEBIH has direct contact with more than 3000 small and medium sized farms in Hungary, therefore the photonic solution can be tested under various field conditions.
Strategic Objective as defined in Photonic Strategic Roadmap
"The European photonics industry must now continue working with the European Commission and national policy makers to coordinate a joint approach to innovation, and to pool investments for enabling the rapid development of new products and minimising times to market. This to market approach needs to include the entire value chain, from advanced research through to technology take-up, pilot lines, and manufacturing platforms."
PHOTONFOOD contribution
The PHOTONFOOD consortium represents the entire "photonics value chain", from advanced research partners (UULM, BOKU, WU, NMBU) to component manufacturers (NP), systems engineering and manufacturing companies (HS, IRIS), laboratory analytics service providers (ROMER), end-users among food businesses (BAMA, BIGH, Seeberger), food control authorities (NEBIH), and provider of large-scale integrated traceability solutions (IBM in advisory role). Customer and stakeholder involvement, prototype development, pilot production, on-site validation, end-user testing, market scanning and further business development can be seamlessly coordinated in the consortium, ensuring minimal time-to-market and rapid scale-up.

1.3 Concept and methodology

1.3.1. PHOTONFOOD concept

A large number of studies have demonstrated that **MIR spectroscopy in combination with advanced data analysis has the capability to identify and detect a broad range of microorganisms and chemical contaminants** in a lab setting[15,16]. Since its invention, the interest from industries in this technique has grown considerably. The technology can be used for identification of contaminants at the genus, species level and in

[15] Shapaval. V., et al. Applied Microbiology, 2013, 114, 788-796, doi:10.1111/jam.12092.
[16] Lecellier. A., et al. Int J of Microbiology, 2014, 3;168-169:32-41.doi: 10.1016/j.ijfoodmicro.2013.10.011.

most cases even down to strain level. For bacteria and, recently for fungi, it has been shown that identification is possible with an accuracy above 90% on species level. For moulds, MIR spectroscopy outcompetes any existing technology in accuracy and the broad range of fungi that can be addressed[16,17]. However, state-of-the-art applications are performed in environments and under conditions that are not relevant for on-site use along the farm-to-fork food production chain[16,17]. State-for-the-art sample preparation protocols for detecting and identifying microbial contamination by MIR analysis are based on culturing and sub-culturing whereas spectra are obtained in pure culture [16,17].

The **detection of mycotoxins by MIR spectroscopy has been demonstrated by the consortium in the spectral range of 1500–1800 cm^{-1} in extracts from grains** (see Fig. 2). However, the challenge is the exceedingly low limits of detection that are required from a legislation point of view. In addition, the potential presence of multiple mycotoxins and contaminants makes the situation even more difficult. The low concentrations of mycotoxin and other contaminants such as antibiotics and pesticides make the direct detection and separation of signals in food matrices impossible. To date, this challenge can only be met by using sophisticated analytical techniques in especially equipped laboratory (e.g., LC-MS/MS) or via antibody-based immunoassays (e.g. ELISA).

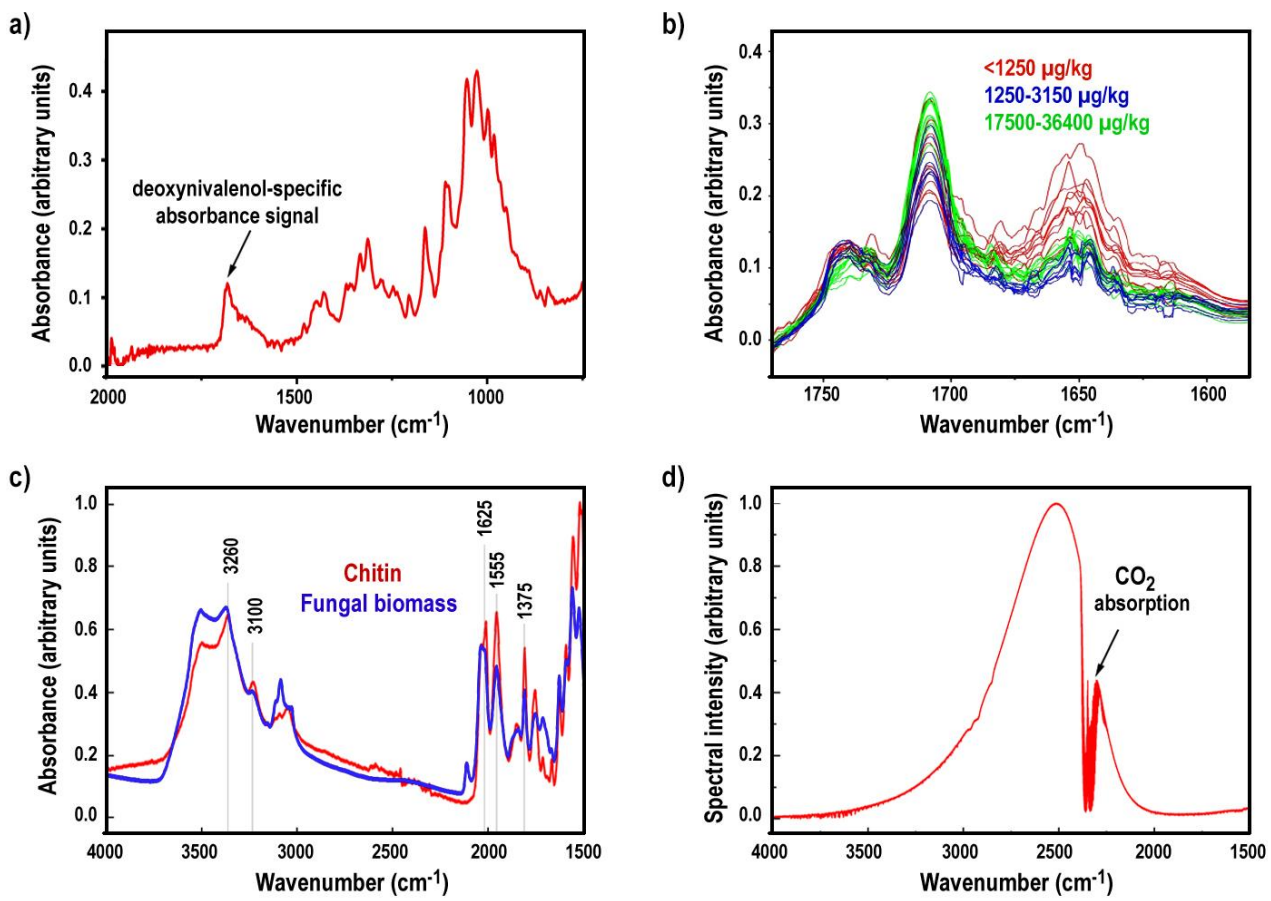

Figure 2: State of the art examples of IR detection of fungal and mycotoxin contaminations. (a) FTIR-ATR spectrum of mycotoxin deoxynivalenol on a paper-based sampling strip. (b) QCL IR spectra of 24 maize extracts with different deoxynivalenol contamination levels (Reprinted from Sieger et al. J Scientific reports 2017, 7:44028). (c) FTIR spectra of chitin and fungal biomass with the characteristic spectral biomarkers for detection of fungal contaminants. (d) Emission spectrum of an IC-LED designed for the 2500 wavenumber (the strong absorption features around 2300 cm^{-1} are caused by CO2 present in the measurement path).

The **PHOTONFOOD concept** is to perform preconcentration of samples using paper-based microfluidics strategies and to combine the cheap and environmentally friendly paper-based microfluidics with label-free MIR photonic read-out devices (see Fig. 3).

While paper microfluidic tests for numerous applications in point-of-care-testing, and environmental research have been demonstrated with detection limits in the mid ppb range[17][18] , few of these offer an integrated solution that also addresses sampling and sample pretreatment. Therefore, the integration of preconcentration

[17] Hariharan *et al.*, *Opt. Mater. (Amst)*, 2019, vol. 94, pp. 305–310.
[18] M. Medina-Sánchez, M. Cadevall, J. Ros, and A. Merkoçi, *Anal. Bioanal. Chem.*, 2015, vol. 407(28), pp. 8445–8449.

would lead to improved detection limits. The WU team has demonstrated such paper microfluidic sample preparation, including paper-based two-phase countercurrent extraction[19], and combining paper and 3D-printed microfluidics for controlled preconcentration[20]. Very recently, WU has developed a miniaturized total-analysis system for the detection of allergens in food products, integrating 3D-printed components with a lateral flow immunoassay, and smartphone readout. This system has been successfully used by an untrained user (15 years old) to differentiate between a cookie containing hazelnut allergen and one without[21]. PHOTONFOOD will built on this experience and develop paper-based microfluidics that can be used by an untrained user such as a consumer and will be tested by end-users as outlined in the stakeholder involvement section on page 29.

In preparation of this proposal, partners UULM, WU and BOKU demonstrated the feasibility of the fundamental concept of combining paper-based sampling strategies with label-free MIR read-out technologies. They have recorded the MIR signature of the exemplary mycotoxin deoxynivalenol (DON) sampled via a paper sampling strip from solution, and then directly detected it via conventional MIR-ATR spectroscopy (Fig. 2a). The experiment shows that high-quality spectra can in principle be obtained from paper-based microfluidics pressed on an ATR surface. PHOTONFOOD will build on this technology and develop 3D printed interface for microfluidics and the optical surfaces of the MI-FI and HI-FI devices.

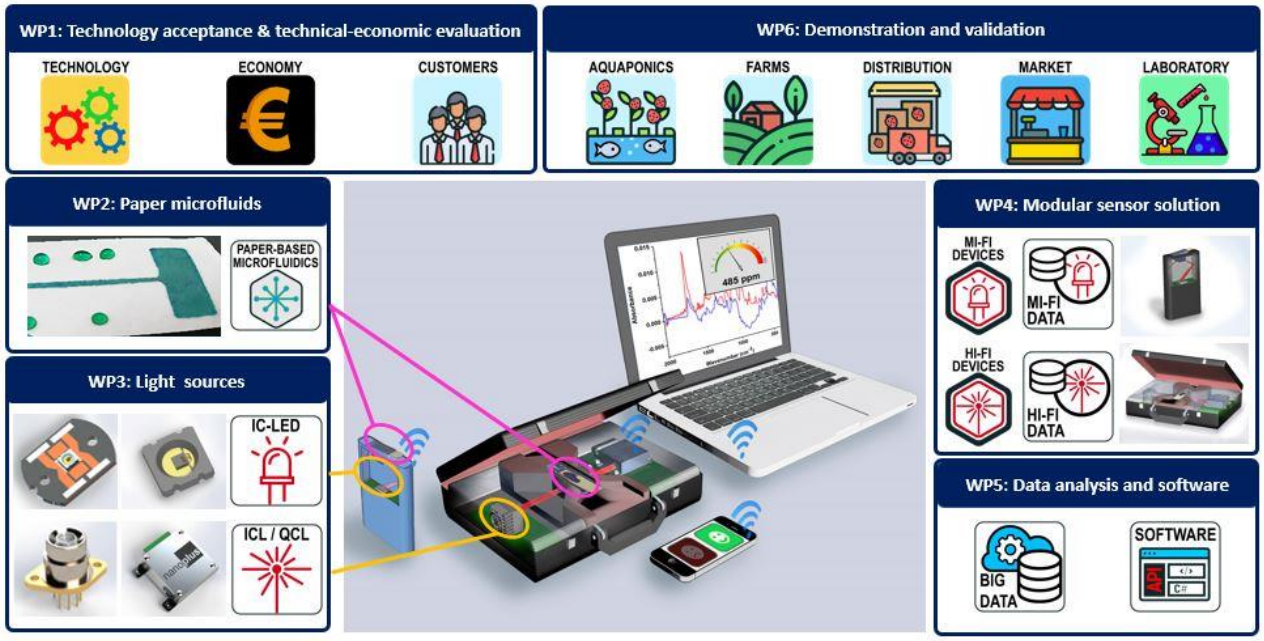

Figure 3: The overall concept of PHOTONFOOD.

Furthermore, the consortium members UULM, BOKU and IRIS have shown that mycotoxins can be detected through molecular signature changes in the spectral region from 1500–1800 cm^{-1} [22] (Fig. 2b). NMBU has shown that filamentous fungi have strong and characteristic molecular signatures in the MIR spectral region[23] (Fig. 2c). Acetylglucosamine is a subunit of chitin that is present in the cell wall of all fungi and that gives rise to strong signals in the amide I and amide II region of the MIR (1500–1700cm^{-1}) (Fig. 2c). Antibiotics and pesticides show discriminative molecular signatures in this spectral range as wel[24]. Therefore, PHOTONFOOD will develop tailor-made light sources for this spectral range. Partner NP has already developed breakthrough IC-LED technology with devices emitting in the 2500-3333 cm^{-1} spectral range with several mW of output power at room temperature (Fig. 2d). Adapting the PHOTONFOOD solution to new contaminants that require other spectral ranges in the MIR will be easily possible in the complete 3333-1000 cm^{-1} range. This will be possible due to the ability to specifically engineer the band structure of ICL and QCL

[19] G. IJ. Salentijn, M. Grajewski, and E. Verpoorte, Lab Chip, 2017, vol. 17(20), pp. 3401–3404.
[20] G. IJ. Salentijn and M. H. Hoekstra, "Sensor Cartridge for chemical assays of a liquid sample containing analyte molecules - Patent No. WO2018226097," 2017
[21] Ross GMS, Filippini D, Nielen MWF, Salentijn GIJ (**2020**), Lab Chip, 2020, under review
[22] Sieger M, et al. *J Scientific reports* 2017, **7**, 44028.
[23] S. Dzurendova, et al., PlosOne, 2020, *accepted.*
[24] Undabeytia T, Morillo E, Maqueda C. J Agric Food Chem. 2002, vol. 50(7), pp. 1918-1921.

gain material as well as to modify the design parameters of the devices developed within PHOTONFOOD in order to cover additional wavelengths. ICLs and IC-LEDs can be operated in continuous-wave mode up to room temperature in this wavelength range with low power consumption, rendering them very cost effective. For QCLs however, power consumption is significantly higher, yet, pulsed operation at room temperature yields significantly higher power output compared to ICL based devices. The research on the photonic devices is based on results obtained in previous FP7 and Horizon 2020 projects such as SensHy, MIRPHAB and MIRACLE (see Table 4) as well as national projects that laid the basis for long wavelength ICL and high performance QCL devices[25,26]. PHOTONFOOD will be using this knowledge as a basis for the dedicated gain material and the chip development for IC-LEDs and widely tunable lasers in the 1500-1800 cm^{-1} spectral region. The photonic devices developed will have TLR of 6 at the end of the project.

Portable solutions that can be used for monitoring and reference analysis along the food chain that allow the detection of microbial and chemical contaminations (e.g., mycotoxins, antibiotics) do not exist and are highly desirable[22,27,28,29] . Therefore, the PHOTONFOOD solution will include a low-cost mid-fidelity (MI-FI) device for monitoring and a high-fidelity (HI-FI) device for reference analysis, with a mobile application for easy, on-site operations The MI-FI data will be of sufficient quality to decide if contamination is present or not, while the HI-FI device will deliver high-quality spectra, providing the ground for the later accreditation of the device for a larger range of contamination problems. These devices will supersede and advance the developed TRL6-level technology established at UULM and partners for the analysis of mycotoxin contaminations in the laboratory[22,28] (see Table 4).

To identify if contamination is present or not and to estimate the level of contamination, powerful machine learning, deep learning and other advanced date mining techniques will be used and wrapped into applications for use on mobile phones and portable computers. At the end of the project period the PHOTONFOOD solution will have been demonstrated and validated for selected contaminants and food commodities. Initially, the focus will be on some of the most important microbial and chemical contaminants in food, i.e. fungi, mycotoxins, antibiotics and pesticides. Furthermore, we will demonstrate that the PHOTONFOOD solution can be easily and broadly adapted to any food contaminant.

The combination of this broad range of ground-breaking technologies requires highly interdisciplinary competence in such fields as MIR spectroscopy, light source and waveguide development, paper-based on-site chemical analysis, food spoilage microbiology, and data science.

The PHOTONFOOD project consists of three essential phases and stakeholders will be involved during all three: (i) The first phase is the technology development phase during which the photonic sensor solution will be developed. (ii) During the second phase the solution will be demonstrated in lab settings. (iii) The third phase includes the demonstration and validation of the platform in real settings at critical control points along the food production chain. During the **first phase**, a technology acceptance and technical-economic evaluation (WP1) will go along with the development of photonic components and data analysis algorithms (WP3-5) and paper-based microfluidics for rapid and reproducible sampling from hard and soft food products and sample pretreatment (WP2). In the **second phase**, the pre-validation of samples obtained from critical control points along the food chain (raw materials, water from the processing environment (aquaponics), final food products) will be done in laboratory settings. In the **third** and final **phase of the project**, the complete solution will be validated in real settings by relevant stakeholders along the food supply value chain. For the validation, one HI-FI and five MI-FI instruments will be produced. The demonstration and validation will be accompanied by evaluation of technology acceptance and technical-economical evaluation in real settings: on farms (wheat and walnut farms), in food processing facilities (dried fruit producers, peanut processing), wholesale and retail (grain, nuts and dried fruits), and novel food production (aquaponics), as well as food laboratories (authority and private laboratory units). The interaction with users along the complete food supply chain will be crucial for addressing user requirements.

1.3.2 PHOTONFOOD main results and technology readiness levels

[25] N. Schäfer; J. Scheuermann; R. Weih; J. Koeth; S. Höfling, Optical Engineering, 2019, vol. 58(11), pp. 117106.
[26] L. Naehle, et al. "Monolithic tunable GaSb-based lasers at 3.3 µm", Electronics Letters, 2011, vol. 47, pp.1092.
[27] K. Roodenko et al. in Proc. SPIE 11233, Optical Fibers and Sensors for Medical Diagnostics and Treatment Applications XX, 112330C (9 March 2020).
[28] T. Öner, P. Thiam, G. Kos, R. Krska, F. Schwenker, B. Mizaikoff, World Mycotoxin J., vol. 12(2), pp. 113 – 122.
[29] McMullin, D., Mizaikoff, B. & Krska, R. Anal Bioanal Chem, 2015, vol. 407, pp. 653-660.

PHOTONFOOD will provide innovative photonic IC-LED and ICL/QCL light sources and waveguides for the MIR to overcome drawbacks of existing photonics technology with respect to price, precision and power and will adapt the solution to specific requirements for on-site measurements in the whole food chain. These advances will provide a portable solution that can be used for photonics monitoring and reference analysis in the whole food chain. The photonics solution will contain a unique sample preconcentration that allows the photonics detection of contaminations at the lowest levels. The photonics solution will be combined with a software application that enables the use on-site (mobile and computer application). The prototyping planned in the project and the demonstration of the solution will pave the way to a commercialization after the project period. Table 3 summarizes the technology levels for the different components developed in the project, while Fig. 4 explains the TRL for the complete solution.

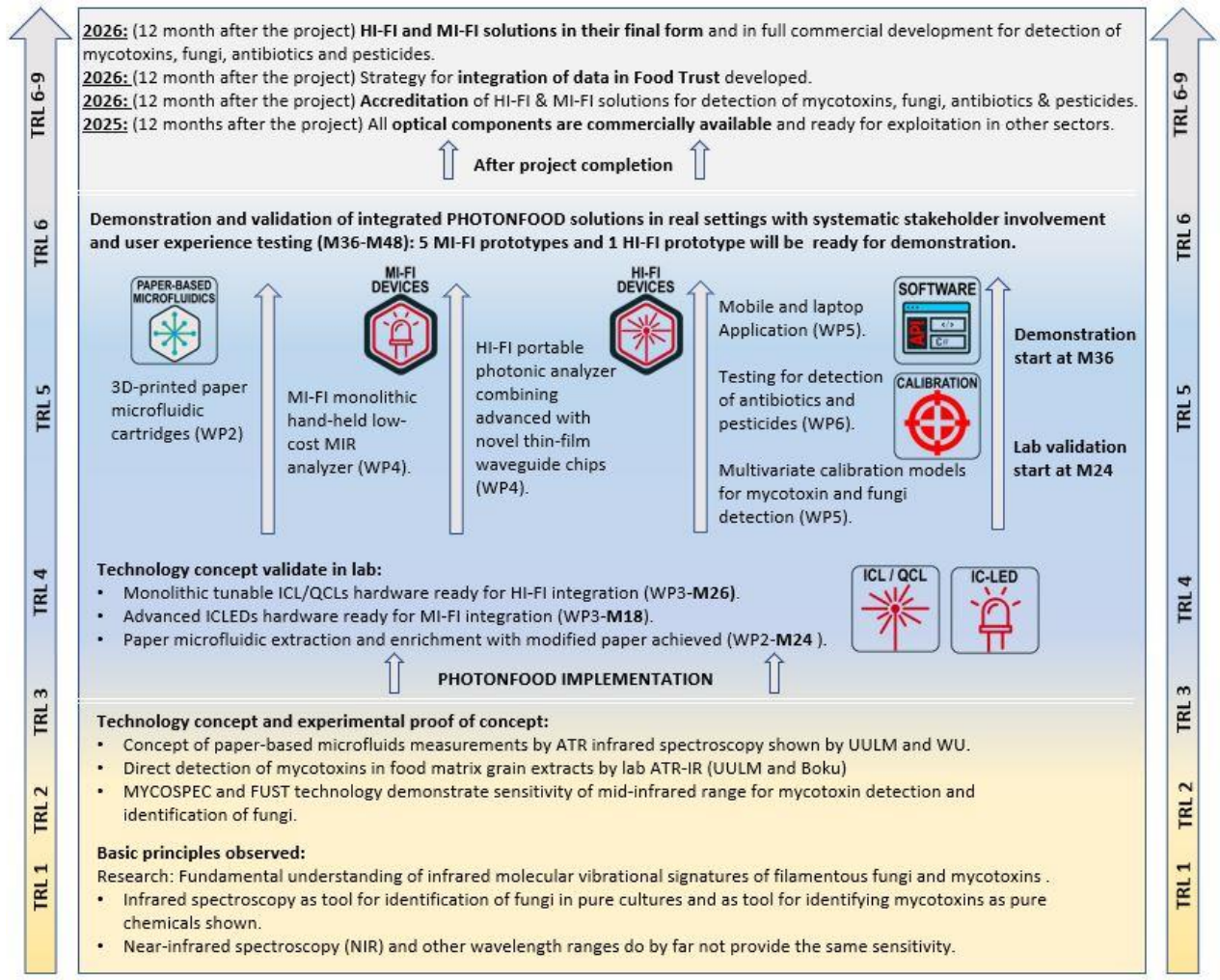

Figure 4: TRLs description for the PHOTONFOOD solution prior, at the start, during, and after the project completion.

Table 3: The individual components of PHOTONFOOD and their TRLs at the project start and after project completion

1. Paper-based microfluidics - preconcentration and separation
TRL 3-4: The proof-of-concept, which is the MIR detection of mycotoxins from a paper strip has been established for this proposal. Proof-of-concept has already been published for functionalities that will be employed in this project, such as paper-based liquid-liquid extraction, as well as modified-paper-based selective enrichment and immunocapture. These approaches will be further developed and tailored towards the mycotoxin applications in PHOTONFOOD.
TRL 5-6: By the end of the project, the most appropriate concept for sample pre-treatment will have been selected, validated and demonstrated in a real scenario. Sampling and pre-treatment will be combined into

easy-to-operate cartridges, which will be made available for testing. Demonstrations of the developed technology will be performed for the selected food safety cases.
2. 3D-printed interface for microfluidics and optical surfaces
TRL 4-5: A 3D-printed micro total analysis system has been developed in the WU labs, which allows for facile extraction and sample pretreatment of allergens from a solid food matrix, followed by a lateral flow immunoassay. This immunoassay has been validated as a screening assay, and the total system was successfully used by an untrained individual.
TRL 5-6: By the end of the project, novel strategies will have been developed and validated/demonstrated in real-life scenarios for the hard and soft food products relevant to PHOTONFOODS. Sampling and pre-treatment will be combined into easy-to-operate cartridges, which will be made available for testing. Also, facile interfacing with MIR readout will have been achieved at this point.
3. HI-FI ICL/QCL light source for 1500-1800 cm^{-1} range spectral range
TRL 4: The technology has already been investigated for the 5000 cm^{-1} wavelength range based on diode lasers and is currently under evaluation for the sensing of blood alcohol levels within the Driver Alcohol Detection System for Safety (DADSS) program by the Automotive Coalition for Traffic Safety (ACTS). The patent for the underlying technology is pending. NP has the exclusive right of the use of the technologies for any application beyond alcohol sensing in cars.
TRL 6: The tunable laser technology will be developed further to extend it to the MIR wavelength range in the 1500-1800 cm^{-1} spectral region. Depending on the performance requirements in terms of tunability and output power either QCL or ICL gain material will be used for the realization of tailor-made lasers for mycotoxin detection within the PHOTONFOOD project. The proposed novel ICL/QCL based light source will provide essential advantages with respect to ruggedness, compactness, speed and costs compared to tunable external cavity systems currently available on the market.
4. MI-FI MIR IC-LED light sources for 1500-1800 cm^{-1} range spectral region
TRL 3-4: TRL4: The general technological principle has already been demonstrated for the 2500-3333 cm^{-1} wavelength range which is the sweet spot for the performance of the ICL technology. The technology has been established for this wavelength region.
TRL 6: A suitable high-performance MIR IC-LED for the 1500-1800 cm^{-1} spectral region based on ICL technology is available for the integration into the MI-FI device. Processing, packaging as well as characterization routes and technologies that allow for low-cost target applications are investigated and evaluated.
5. HI-FI MIR portable photonic device in the 1500-1800 cm^{-1} range
TRL 3-4: UULM and partners have pioneered mycotoxin sensors based on tunable QCLs and thin-film waveguides in a compact format and have readily demonstrated its fundamental utility for the detection of mycotoxin contamination in wheat, corn, and peanuts[22] (see Table 4). In order to develop high-fidelity (HI-FI) portable MIR spectroscopic devices providing the level of sensitivity and selectivity needed for the detection of relevant fungal and mycotoxin contamination at defined detection limits in food/commodity matrices, the brightest possible MIR light sources are required.
TRL 6: One high-end portable (i.e. suitcase-style) quantum cascade laser-based HI-FI analyser platform based on thin-film nano crystalline diamond waveguides as a sensing interface will be the reference system for providing MIR molecular signatures of commodity matrix changes and contaminants with maximum spectral resolution and sensitivity.
6. MI-FI MIR handheld photonic device for the 1500-1800 cm^{-1} range
TRL 3: The fundamental concept of low-cost IC-LED-based handheld analyzers with limited spectral resolution has successfully been shown for near-infrared devices (see for example Consumer Physics SCIO technology). The proposed mid-fidelity (MI-FI) device will use similar optical concepts, yet translated into the MIR spectral domain with appropriate light source (IC-LED) and detector technology augmented by a single-bounce diamond attenuated total reflection sensing interface.
TRL 6: Five field-deployable prototypes of hand-held low-cost monolithic dispersive MIR analyser systems based on IC-LEDs for obtaining rapid medium-resolution MIR molecular signatures will be provided and tested at small-to-medium farms or supermarkets. Each device will fit into the palm of a hand, will be controlled via a smartphone app, and will be compatible with the paper microfluidic sampling strategy.
7. Advanced data analysis calibrations

TRL 4: The use of calibrations based on MIR spectral molecular signatures for the detection of various contaminants has been successfully shown in a lab environment. Consortium members have in on-going projects developed successfully preprocessing and calibration models based on ATR MIR spectral data with narrow spectral ranges.
TRL 6: Pre-processing for MI-FI and HI-FI waveguide MIR data will be developed for a short spectral range 1500-1800 cm^{-1}. Calibration models for estimation of levels of contaminants (HI-FI data) and for monitoring (MI-FI data) will be established for fungi, mycotoxins, antibiotics and pesticides. The models will be ready to be implemented in software applications for mobile phones and portable computers.
8. PHOTONFOOD software application for mobile and laptop
TRL 3-4: A software and user interface template for ATR MIR data is available from the MIRACLE project (see Table 4) where a MIR arthroscopy innovative imaging system for real-time clinical in-depth examination and diagnosis of degenerative joint diseases is developed.
TRL 6: PHOTONFOOD will produce a mobile and a laptop application for the MI-FI and HI-FI contamination management solutions, providing the digitalization of the data, the integration of the models developed in the project and a friendly user-interface.

1.3.3 Projects preceding PHOTONFOOD

Table 4: List of projects preceding PHOTONFOOD and their relevance for PHOTONFOOD

Project name, funding programme (period)	Relevance for the proposed project [involved partners]
FUST - Source tracking and monitoring of mould contamination in food production (FP7-SME-2012-315271-FUST) (2012-2014)	Development of automated lab system for source tracking of microorganisms along the food production chain **[NMBU]**. The system will be used for spectroscopy benchmarking for fungal contamination in PHOTONFOOD.
MYCOSPEC - Novel iInfrared Spectroscopic tools for Mycotoxin Ddetermination in MYTOOLBOX, H2020 SFS, GA 678012, Safe Food and Feed through an Integrated ToolBox for Mycotoxin Management	QLCs and thin-film GaAs/AlGaAs waveguide technology, **[UULM, BOKU, IRIS]**. The waveguide technology will be used for benchmarking for mycotoxin contamination in PHOTONFOOD. Providing high-quality and state-of-the-art LC-MS/MS based reference analysis for the determination of all regulated mycotoxins and beyond, including deoxynivalenol (DON) in wheat, and total aflatoxin (B1, B2, G1, G2) contamination in nuts.
DeepHyperSpec - Combining spectral and image information in the analysis of hyperspectral imaging data, acronym (Norwegian Research Council grant Nr. 289518) (2019-2022)	Development of deep learning techniques for mining molecular spectral signatures **[NMBU]**. The developed advanced data analysis techniques will be used for calibration establishment in PHOTONFOOD.
MIRACLE - MIR arthroscopy innovative imaging system for real-time clinical in-depth examination and diagnosis of degenerative joint diseases (H2020, ICT-30-2017) (2017-2021)	Tailored data analysis tools and software for narrow wavelength ranges available from the tailored laser light sources of PHOTONFOOD **[UULM, NMBU, IRIS, NP].** Knowhow for calibration and spectral pre-treatment methods for short wavelength regions.
MOLOKO – Multiplex photonic sensor for plasmonic-based Online detection of contaminants in milk, H2020-ICT-2017-1 No. 780839 (2018 –2021)	Manufacturing, implementation and validation of a self-managing and automatic miniaturized integrated photonic sensor solution. **[NEBIH]** Validation experience and method will be used in WP6's validation process.
INVITE - Innovations in plant variety testing in Europe to foster the introduction of new varieties better adapted to varying biotic and abiotic conditions and to more sustainable crop management practices, H2020-SFS-2018-2 No. 817970 (2019 – 2024)	Sustainable crop management and practice. **[WU, NEBIH]** The experience on crop waste reduction has been integrated into the theory of PHOTONFOOD project.
SafeConsume - Safer food through changed consumer behaviour: Effective tools and products, communication strategies, education and a food safety policy reducing health burden from foodborne illnesses, H2020-SFS-2016-2 No. 727580 (2017-2022)	Development of food safety risk-related tools, education and policy addressing EFSA and EC. [**NEBIH]** The expert network and the achieved social capital of the SafeConsume project will be exploited for the promotion of the innovative photonic system.
SCHEMA - MIR liquid phase sensing technologies for the detection of trace chemical contaminants in water in real-world application scenarios (#614002, 2013-2017)	Chemical MIR sensing technology based on fibreoptic waveguides providing know-how on optically efficient MIR-based chem/bio sensors, providing fundamental optical concepts for PHOTONFOOD **[UULM]**
TROPSENSE - MIR laser-based (QCL, ICL) sensing technologies for detecting tropical diseases in exhaled breath (#645758, 2014-2017)	Advanced MIR laser technology for clinical/medical diagnostics providing knowledge on miniaturization of MIR sensing concepts for PHOTONFOOD **[UULM]**

APOSEMA - MIR laser-based (ICL) sensing technologies for exhaled breath analysis, acronym (#645758, 2014-2017)	Next-generation ICL technology for MIR sensing applications providing key technologies for robust in-field usage facilitating PHOTONFOOD **[UULM, NP]**
VOGAS - MIR sensors combined with eNoses for the detection of gastric cancer (#824986, 2019-2022)	Combined MIR-eNose sensing systems for clinical/medical diagnostics providing fundamental concepts on MIR system integration, further evolved in PHOTONFOOD **[UULM]**
BREAK BIOFILMS - MIR waveguide-based analyser platforms for the molecular analysis of biofilms, biofilm formation, and biofilm prevention (#813439, 2019-2022)	Advanced MIR sensing platforms for the analysis of microorganisms providing fundamental experience on the MIR signatures of bio-matrices, also relevant in PHOTONFOOD. Additionally, alteration of biofilm and bio-matrix MIR-signatures upon exposure to e.g. spoilage inducing microorganisms, disinfectants or environmental influences is investigated. **[UULM]**
SENSHY - Research on novel gas sensors for hydrocarbons in the 3-3.6µm wavelength range. (EU FP7-ICT-2007-2, No.223998)	Research and Development of ICL lasers. Project results lead to the first commercialization of DFB ICL lasers for sensing and the project results formed the technological basis for ICLs a NP. The project formed the basis of all research and development at NP with respect to its ICL technology. **[NP]**
MIRPHAB- European pilot line for single access point to all the MIR technology and components to build customized chemical sensors (H2020-ICT-28-2015, No 688265)	Maturation and qualification of ICL technology. Improving performance of long wavelength ICLs in the 2000 cm^{-1} spectral region. Results with respect to performance improvements on the epitaxy in the 2000-1600 cm^{-1} spectral range can be used a starting point for the development of the IC-LED and ICL gain structures required for the PHOTONFOOD project. **[NP]**
Mytoolbox - Safe food and feed through an integrated toolbox for mycotoxin management (H2020, SFS-13-2015) (2020-2024)	Infrastructure for the web-based MyToolBox cloud platform for control and reduction of mycotoxins **[BOKU, WU, IRIS]**
Cloud-enabled infrared spectroscopy platform for the rapid integration and customisation of affordable, real-time chemical monitoring solutions for food and pharmaceutical industries, acronym 'VISUM' (H2020, SMEInst-07-2016-2017)	IR spectroscopy cloud-enabled platform for the rapid, easy integration and customization of real-time monitoring systems for improved process control, product quality and safety in the food and pharmaceutical industries. The experience achieved in this project in relation to the development of monitoring solutions and cloud platforms connectivity will be a reference for the software platform to be developed in PHOTONFOOD **[IRIS]**
HYPERA - Affordable Hyperspectral Imaging System for on-line foreign body detection and chemical composition analysis, (H2020, SMEInst-07-2016-2017)	Development of a novel and disruptive combination of Shortwave Infrared Hyperspectral Imaging in reflectance with Near Infrared Multispectral Imaging in reflectance/transmittance and chemometric software for on-line foreign body detection and chemical composition analysis. The experience regarding chemometric software integration will be a reference for the software platform to be developed in PHOTONFOOD **[IRIS]**
MULTICOOP - Multidisciplinary approach to strengthen and establish novel platforms for the comprehensive assessment of food and feed safety (H2020-TWINN-2015, 2014-2017)	Providing high-quality and state-of-the-art LC-MS/MS based reference analysis for the determination of total deoxynivalenol (including acetylated and masked forms) contamination in wheat samples. Experience will be exploited in the benchmark process of innovative photonic device with reference methods of WP6 **[BOKU]**
MYCORED - Novel integrated strategies for worldwide mycotoxin reduction in food and feed chains (FP7-KBBE-2007-2AFP 7, 2009-2014)	Organization of a miniaturized ring trial with a minimum number of 3 participants. The ring trial experience and method will be used in WP6's validation process **[BOKU]**
FoodSmartphone - Smartphone analyzers for on-site testing of food quality and safety (H2020-MSCA-ITN, grant 720325, 2017-2020).	Development of simplified tools for on-site sample preparation by non-experts **[WU]**. The outcomes of this project directly feed into the development of on-site sample pre-treatment for WP2 in PHOTONFOOD.
Citizen Science Innovation Initiative (Jointly funded by province and Regional and European Fund for Regional Development (EFRD), acronym Proeftuin CSII (2018-2020).	Design and validation of portable analytical instruments and devices in food and health sciences **[WU]**. The validation and handheld spectroscopy expertise gained from this project will be beneficial to PHOTONFOOD in designing user-friendly solutions for on-site food testing.

1.3.4 Methodology

The PHOTONFOOD project is based on four technical WPs, a WP on user acceptance, and a WP on demonstration and validation. The interaction of the WPs is illustrated in Fig. 5.

In **WP1**, stakeholder and user expectations will be obtained, analyzed and implemented in order to adapt the PHOTONFOOD technology during the development process to the needs of the end user. The PHOTONFOOD solution needs to be tailored to the needs of a broad range of users including food laboratories (private and authority units), food business organizations from different segments of the value chain, farmers and end-users.

WP1 will identify stakeholders from agriculture, food processing, retail, public food control and end-users. In WP1, a preliminary requirements and expectation survey will be conducted among stakeholders, followed by the organization of technical experience collection during the validation phase. Later in the project, the demonstration process will take place (WP6), in which prototype devices will be deployed by project partners to assess their performance under operational conditions. User experiences will be collected in a systematic way, according to the elements of the UTAUT model (see Fig. 6). The UTAUT model allows us to predict the willingness to use of the novel technology and device through the factors that influence behaviour. Parallel to that, based on the analysis of cost elements recorded and assessed during the demonstration phase, a comparative unit cost assessment will be conducted against the currently available reference methods. After all necessary input gathered for the UTAUT model, a business plan will be composed to foster further development of the technology. WP1 activities assure that the final product will be reflecting on stakeholders' expectations and will meet a high level of market acceptance. Without such a thorough analysis and implementation of the feedbacks, the product would not be successful and profitable in long term.

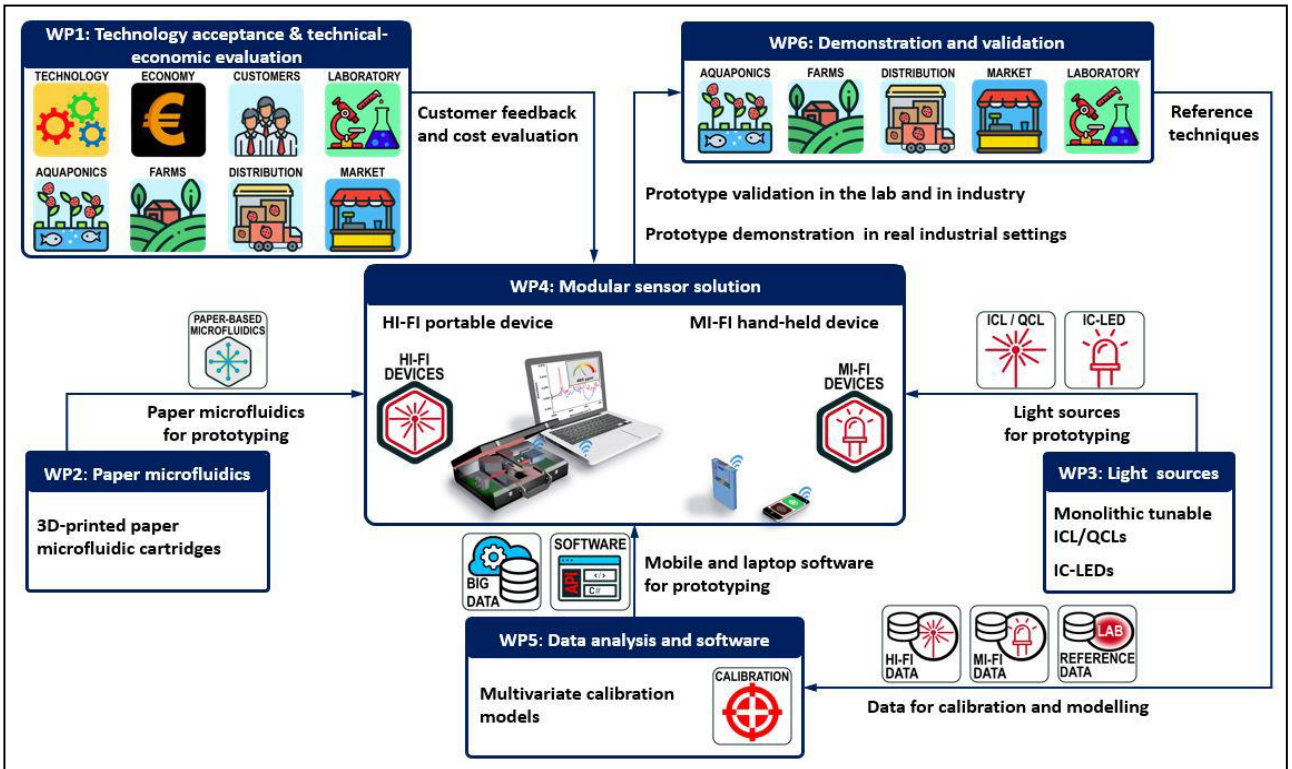

Figure. 5: Overview of the PHOTONFOOD methodology. The overview shows the relation between the technical WPs, the evaluation of the technology acceptance, the technical-economical evaluation, and the demonstration and validation.

In **WP2** an integrated workflow will be developed on a paper microfluidic device from sampling, to clean up of complex food samples, and enrichment of the compounds of interest prior to read-out by the HI-FI and MI-FI devices prototyped in WP4. Paper micro-fluidic device will require only an aliquot of a solid food sample, which is then automatically grinded into small particles, extracted with solvent, and loaded onto a smart paper device for separation and/or preconcentration. Routes to be explored include on-paper liquid-liquid extraction, and immunocapture. Moreover, the use of surface chemistry will be explored to modify the cellulose fibres in paper to allow selective extraction of compounds based on their physicochemical properties. Afterwards, the microfluidic device will be coupled to either the HI-FI or the MI-FI photonic sensing devices developed in WP3 for measurement of the MIR molecular signatures characterizing the level of microbial and chemical contamination. To ensure maximal compatibility between modules, 3D-printing will be employed for the development of a protective user-friendly cartridges with a custom interface to the waveguide.

WP3 is dedicated to the development of next-generation MIR light sources for medium-end (i.e., IC-LED) to high-end (i.e. ICL/QCL) sensor systems for the portable PHOTONFOOD prototypes provided by WP4. Previous studies by partners of this consortium (UULM, BOKU, IRIS with the EU-project MYCOSPEC) have readily demonstrated that fungal/mycotoxin contamination can be potentially detected via specific molecular signatures in the 1500-1800 cm^{-1} µm MIR spectral window. With the development of advanced MIR LED and laser light sources operating in this spectral range, a photonic sensing platform technology is made available

for applications extending above and beyond the food quality/food safety scenarios due its scalability resulting out of the low costs of the IC-LED source. Two major light source developments will be pursued: (a) IC-LEDs emitting in the 1500-1800 cm^{-1} spectral range and (b) monolithical widely tunable cascaded lasers based on either interband or quantum cascade gain material, respectively, in the 1500-1800 cm^{-1} region. Partner NP has already developed breakthrough IC-LED technology with devices emitting in the 2500-3333 cm^{-1} spectral range with several mW of output power at room temperature, which have been tested according to their analytical performance with preliminary studies by partner UULM (Fig. 2d). During the validation in WP6, both IC-LEDs (in the 1500-1800 cm^{-1} and in the 2500-3333 cm^{-1} spectral range)will be evaluated for their ability to reveal spectral molecular signatures of contaminants considered in the PHOTONFOOD project (fungi, mycotoxins, antibiotics, pesticides).

WP4: The golden standard for the detection of contaminants via MIR spectral molecular signatures is currently represented by broadband MIR spectra obtained by high-end laboratory equipment. In the project, the detection of mycotoxins is done using HI-FI and MI-FI solutions with a considerably shorter wavelength range. The light sources developed in WP3 will be integrated into (prototype) devices serving different applications (monitoring and reference analysis) with appropriate high-fidelity to medium-fidelity information. HI-FI devices will allow to estimate contamination levels, while MI-FI devices will provide data that is sufficient for monitoring. Partner UULM together with BOKU and IRIS have developed an innovative thin-film waveguide technology in a previous project (EU-project MYCOSPEC, see Table 4) with the fundamental sensing interface being commercialized (see MIR-Guide™; https://www.mgopticalsolutions.com/en/manufacturer/mg-optical-solutions/). The innovation of the present project is surpassing the thin-film waveguide made from semiconductors with the breakthrough technology of nanocrystalline diamond (NCD), which has demonstrated superior properties including biocompatibility and inertness. during pioneering studies by partner UULM (BMBF project GRA//DIA). Two major analyzer/sensor developments will be pursued: (a) Low-cost hand-held MI-FI analyzer combining MIR LEDs developed in WP3 with a conventional diamond ATR element as sampling interface, a grating for dispersing the MIR radiation emitted by the LED, and a line array detector recording wavelength-resolved information with medium spectral resolution. (b) Portable shoe-box-sized HI-FI analyzer providing superior sensitivity by combining the monolithic tunable ICL/QCL light sources developed in WP3 with advanced thin-film nanocrystalline diamond (NCD) waveguides and a single-element MCT detector recording wavelength-resolved information with high spectral resolution.

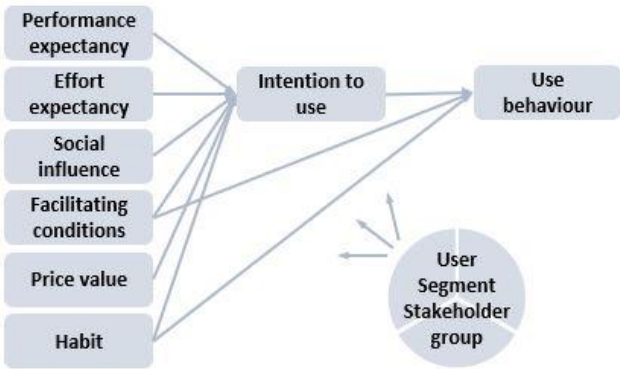

Figure 6: Framework for technology acceptance: UTAUT model = Unified theory of acceptance and use of technology (Venkatesh et al. 2003)

In **WP5** data pre-processing problems will be addressed that are specific for the light sources that are developed in WP3 and the photonic devices that are developed in WP4. Short wavelength ranges pose special challenges for the standardization of spectral data, usually the first step of the data processing of spectroscopic data. The problem will be addressed by following two routes, namely building on model-based pre-processing, an approach developed by the NMBU group for data where only few wavenumbers are available. In addition, approaches from the machine learning domain such as data augmentation will be evaluated. For the development of robust calibration models, augmentation of samples sets will be performed experimentally and by modelling. Calibration models will be established using BioSpec toolbox of machine learning techniques involving a broad range of methods such as Partial Least Squares Discriminant Analysis (PLS-DA), Random Forests (RF), Support Vector Machines (SVM), Artificial Neural Networks (ANN) and Deep Convolutional Neural Networks (DCNN). Two types of models will be established 1) classification models to detect whether the contamination is present or not, 2) classification models to detect the level of contamination. The developed pre-processing and classification and calibration models will provide a platform for further adaption of the PHOTONFOOD technology beyond the project. Furthermore, WP5 will provide a mobile and a computer interface for the prototypes developed in WP4. Already during the project, a strategy will be developed for

how the MI-FI and HI-FI data can be connected to platforms such as the Food Trust™ for utilization of the big data collected over time for improving food safety and reducing waste.

WP6 is dedicated to the validation of the PHOTONFOOD technology. From the beginning of the project, lab-based instrumentation will be used in **WP6** for benchmarking and data collection. This data will be used to make early progress with identifying molecular spectral signatures, data pre-processing and development of algorithms for fungal, mycotoxin, antibiotics and pesticide detection. This will be done in parallel to the development of the photonic components. The **validation has two main phases. The first validation period** will take place after the MI-FI instrument is ready (**after M24**). Whereas PHOTONFOOD partner NEBIH will focus on HPLC-UV/FLD reference measurements, BOKU will provide more accurate and complete information on the range of metabolites in the samples. Since the project requires samples for testing, validation and demonstration during the whole period of the project, the project will organise a preliminary comparability study between NEBIH, ROMER and BOKU to ensure comparability of measurement results by involving HPLC-UV/FLD, LC-MS/MS, ELISA and the photonics devices to be validated. The **second** phase will be in the field and will serve as the **actual demonstration phase at the end user´s level after M36.** This exercise will be organised on site at the end user level, i.e. at the farmer´s place or official testing labs. As the basis for the demonstration and validation exercise (WP6), NEBIH will provide samples collected from the field in Hungary from farms, food processors and retailers, i.e. a set of at least 30 naturally contaminated samples at different contamination levels for each of the investigated commodities. In cooperation with partner BOKU, NEBIH will also provide reference analyses for the collected samples. NEBIH analysed 4185 samples in 2018 and 5699 samples in 2019 using HPLC-UV/FLD and ELISA, respectively. They will thus also be able to provide PHOTONFOOD with both samples and reference data needed to test and validate the developed photonics-based devices both in the lab and on site. As a continuation of the preliminary comparability study, BOKU will organise a ring trial (interlaboratory comparison study) with NEBIH, ROMER, NMBU and UULM.

1.3.5 Gender equality and gender dimension

PHOTONFOOD consortium involves a significant number of female scientists at different level of contributions – WP leader, team researchers, project administration and coordination (see Section 4). Each project partner is committed to the highest standards of ethics and acts according to the national and European regulations (Directive 2006/54/EC). We all declare that everybody is treated equally and respectfully regardless of nationality, ethnicity, sex and gender, similar to the ongoing good practices in our organizations. When hiring new personnel, the decisions are solely based on scientific/technical ground, i.e. the expertise of the person and suitability for the devoted project tasks serve the only bases of any selections.

The research and development activities in PHOTONFOOD will be heavily related to the engineering, photonics innovations and data science, where there is a strong tendency towards male specialists and there is a lack of female representation, both in industry and academia. Thus, PHOTONFOOD project will promote higher level of equality for both male and female specialists by encouraging female students and researchers to engage with the project activities. In addition, PHOTONFOOD project it will interact with the specialists from chemical/microbiological laboratories and food industry where female specialists could potentially dominate, that will positively affect the gender balance promoted in the project. A member of the coordination team, Stine Brekke Vennemo will be appointed as a gender issue coordinator (GIC) who will monitor matters arising in a relation to gender.

1.4 Ambition

Ground-breaking nature of concept and approach: The PHOTONFOOD advances beyond the state of the art are summarized in Fig. 7. The PHOTONFOOD project will develop and demonstrate a photonic solution for monitoring and reference analysis of chemical and microbial food contamination. Today, monitoring and accredited analysis is done in reference laboratories such as represented by our partner NEBIH. On-site monitoring and reference analysis are not possible today. Moreover, there is no accredited reference analysis for on-site food analysis today. PHOTONFOOD will perform ground-breaking research and develop the most advanced photonic devices for precise chemical analysis in the chemically most sensitive region of the electromagnetic spectrum and provide a handheld solution for detecting chemical and microbial contamination in the food chain.

To offer a maximum of flexibility of the solution, PHOTONFOOD will provide two devices. (i) A cheap solution for on-site monitoring in daily routine work. The MI-FI device will have the size of a mobile phone and will be a handheld low-cost (2-4k€) medium-precision/resolution MIR photonic solutions for the detection of contaminants in food matrices. Today, no handheld MIR spectrometer in the size of a mobile phone or comparable exists. While portable device for the near-infrared sensing with limited spectral resolution is offered already by Consumer Physics (see section 1.4.1.6), this spectral region does by far not provide the detection level of the contamination and the discriminative ability of the contaminants that are required for food analysis. (ii) PHOTONFOOD will provide a portable HI-FI device delivering a spectral quality that will be sufficient for accrediting the technology after the PHOTONFOOD project for detecting chemical and microbial contamination in food. The HI-FI device will combine advanced MIR ICL/QCL laser light sources with novel thin-film waveguide chips for label-free detection of chemical and microbial contaminations in food sampled via low-cost paper-based sample handling. Up to know, no solution exists that can be broadly applied for performing reference analysis of food contaminants on site. The advantage of MIR spectroscopic methods is that they can be broadly adapted to many microorganisms and chemical contaminants. MIR spectroscopy delivers a precise chemical fingerprint of any chemical and biological sample that can be presented to the measurement zone of a MIR waveguide. In order to empower spectroscopic methods to perform analysis of the tiniest amounts of contaminants, PHOTONFOOD will provide a ground-breaking paper-based microfluidics technology for sample preconcentration and separation that will be adopted to measurements of the photonic waveguide technologies developed in the project. Today, sampling, extraction, purification and enrichment are all done manually and by trained personnel. Current "portable" solutions still rely on lab-based equipment, and liquid extracts. There is no "easy" way to get purified contaminants from a complex, solid sample. PHOTONFOOD will provide the first miniaturized handheld device that allows non-experts to process food samples and test for trace chemical and microbial contamination through MIR spectroscopy. The PHOTONFOOD solution combines paper-based sample preparation and preconcentration and opens thereby for a broad spectrum of applications of spectroscopy in cases where today the level of contamination was by far too low to allow spectroscopic detection in the food matrix.

	Paper microfluidics and 3D-printed interface	Tunable lasers in the infrared	Low-cost infrared ICLEDs	HI-FI QCL-based photonic analyzer	MI-FI LED-based photonic analyzer	Mobile and Laptop software application
PHOTONFOOD: Beyond State of the art	First miniaturized paper-based Total Analysis System that allows non-experts to process food samples and test for trace contamination through IR spectroscopy	Monolithic widely tunable laser based on cascaded laser structures available for the 1500-1800cm-1 range for high precision measurements in the HI-FI device	Novel integrated MIR ICLED for with broad wavelength coverage in the 1500-1800cm-1 range with 300cm-1 wavelength coverage (1µm spectral width) for cost effective on side measurements in the MI-FI device	High-precision portable photonic analyzer (HI-FI) combining advanced MIR QCL laser light sources with novel thin-film waveguide chips for label-free detection of contaminations in food samples via low-cost paper-based strategies	Medium-precision monolithic hand-held low-cost MIR analyzer (MI-FI) based on novel MIR LED light sources for label-free detection of contaminations in food sampled via low-cost paper-based strategies	Web-accessible platform with advance machine learning models to provide support decision tools for contamination management
Gap	Current "portable" solutions still rely on lab-based equipment, and liquid extracts. There is no 'easy' way to get purified contaminants from a complex, solid sample	No suitable monolithic cascaded laser available for integration into the HI-FI device	No ICLEDs for the 1500-1800cm^{-1} range exists	Portable high-precision/resolution MIR photonic solutions for the detection of contaminants in food matrices do not exist	Hand-held low-cost (2-4 kEuro) medium-precision/resolution MIR photonic solutions for the detection of contaminants in food matrices do not exist	No specific calibration models and data pretreatment tools for MI-FI and HI-FI solutions exist. No ML algorithm running. No DSS engine implemented
State of the art	Modular and proof-of concept paper microfluidic and 3D printed solutions for on-site analytical chemistry, including sample preconcentration and read-out	Distributed Feedback (DFB) QCLs with limited turnability. External cavity (EC-QCLs)	High performance ICLEDs in only the 3333-2500 cm-1 range available and with 300-500 cm-1 (0.5 µm spectral width) coverage of spectral range	MIR spectroscopic equipment that is bound to a laboratory environment	Near-infrared (NIR) low-cost analyzers with extremely limited spectral resolution and molecular discrimination	Software platform (MIRACLE project) 7 QCL-laser wavelength-based estimation and diagnosis of degenerative joint diseases

Figure 7: Summary of the PHOTONFOOD main contributions beyond the state of the art.

For the MI-FI device low-cost powerful light sources, such as LEDs, in the MIR fingerprint region are needed. Today, high performance IC-LEDs are only available for the ranges 2500-3333 cm^{-1}. For detection of chemical and microbial contaminants in food other ranges in the fingerprint region (700-1800 cm^{-1}) of the MIR spectrum need to be covered. The PHOTONFOOD project will develop novel integrated MIR IC-LEDs with 300 cm^{-1} wavelength coverage (1µm spectral width) in the 1500-1800 cm^{-1} range for cost-effective on-site measurements with the MI-FI device. The HI-FI device requires excellent light sources to obtain a high-quality spectra. Portable high-precision/high-resolution MIR photonic solutions for the detection of contaminants in food matrices do not exist. Today, only distributed feedback (DFB) ICL/QCLs with limited turnability and

external cavity (EC-QCLs) are available. PHOTONFOOD will develop the brightest monolithic widely tunable lasers based on cascaded laser structures for the 1500-1800 cm^{-1} range for high precision measurements with the HI-FI device. Model-building and accreditation for the HI-FI technology will be necessary for each contaminant separately, since spectroscopic methods are based on calibration. For prototyping and demonstration, PHOTONFOOD will develop a software platform exploiting the experience from the software platform established in the MIRACLE project (see Table 4). The MIRACLE software platform is developed for a probe utilizing seven QCL-laser wavelengths in the MIR for diagnosis of degenerative joint diseases. Since the portable PHOTONFOOD solution will cover a narrow MIR region, current calibration approaches cannot be transferred without further modification. Today, no specific calibration models and data pre-treatment tools for MI-FI and HI-FI solutions exist. PHOTONFOOD will use the most advanced data analysis approaches and develop them further to address specifically the problem of narrow spectral ranges that will be required for spectral standardization and model building. A web-accessible platform with advanced machine learning models will be implemented to provide support decision tools for contamination management.

1.4.1 State of the art and beyond state of the art

1.4.1.1 Paper microfluidic devices and 3D-printed interface

State of the art

The field of paper microfluidics, which is based on decades worth of knowledge about paper chromatography and other applications of porous cellulose[30], has grown rapidly since the original Whitesides paper from 2007[31]. In this field, paper is used as the substrate for the development of miniaturized total analysis systems (µTAS), which is a concept that goes back to the late 80's. The reason why this branch of microfluidics opted for the use of paper, rather than high-quality chips made in a clean room, is simple: there is an urgent need for affordable and deployable analytical solutions[32],[33]. The high availability, low-cost and superior portability of such paper-based devices make them ideal choices for easy, quick and on-site sample treatment and analysis. Moreover, paper can be readily used for sampling directly from surfaces and liquids alike.

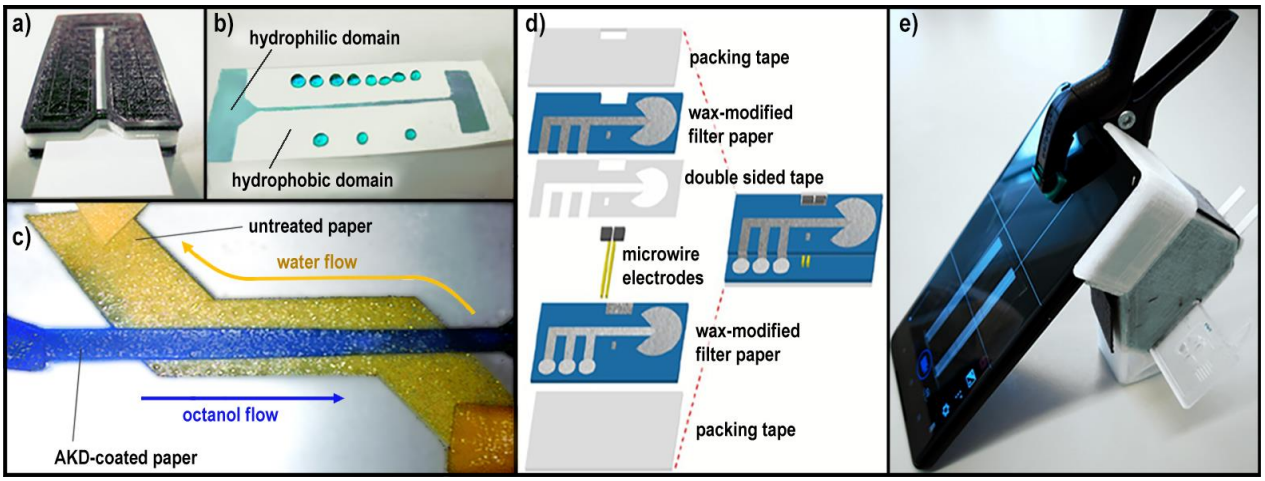

Figure 8: (a) 3D-printed polymer scaffolding used for the fabrication of alkyl ketene dimer (AKD) patterned paper microfluidic device[42]. (b) Paper device with untreated (hydrophilic) and AKD-coated (hydrophobic) domains. An aqueous blue dye solution was applied to both domains to demonstrate wettability by water, creating wetted hydrophilic domain and droplets on top of the hydrophobic domain[42]. (c) Demonstration of 3D paper microfluidic device with two-phase countercurrent flow of water (with hydrophilic yellow dye) and octanol (with hydrophobic blue dye)[43]. (d) Illustration of 3D paper network with integrated electrode sensor[44]. (e) Fully integrated 3D-printed system, containing 3D printed microfluidics and paper-based testing (to be published soon).

In paper microfluidics, liquids are passively moved by capillary action[34], and guided by application of hydrophobic patterns on paper, to confine movement of fluids (Figs. 8a and 8b). The combination of hydrophobic patterning and the stacking of different layers of paper (3D paper microfluidics) enables researchers to design and develop compact systems with a multitude of branching channels through which

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fluids are moved without the need of pumps[35] (Fig 8c). Moreover, through the integration of electrodes, plasmonic nanoparticles, and other modifications, a plethora of detection schemes can be applied to such devices (Figs. 8d and 8e). Recently, the field has expanded towards the implementation of sophisticated device functionality, and proof-of-concepts of on paper-based liquid-liquid extraction[19], on-paper preconcentration[36] and several types of flow control mechanisms have been published[37]. However, those functionalities are modules that remain to be integrated in fully operational systems and need further development to allow (semi) automated operation in a user-friendly device that can be used in an on-site setting.

Paper microfluidic devices can accommodate many functionalities and even include a framework for read-out/detection. However, they lack physical strength, and a user-friendly interface. Thus, 3D printing has been applied to create cartridges, adapters and even fluidic devices via rapid prototyping, thereby revolutionizing the way and speed with which device development is carried out[38],[39]. Different types of 3D printing have been employed, but the main strategy seems to be using fused deposition modelling for larger components, such as casings and adaptors, whereas techniques with higher resolution, such as stereolithography are used for making more functional components, such as fluidic channels and smaller parts. The combination of paper microfluidics with 3D-printing is very powerful[40], as it combines the broad range of functionalities and possibilities from the first, with physical strength, and adaptability of the latter.

PHOTONFOOD going beyond the state of the art

Paper microfluidics is a field with many possibilities and aspects to be explored. Applying multi-layered approaches (3D paper microfluidics) and building 3D-printed cartridges to house, protect and add functionality to such devices is especially interesting[41] (see Fig. 8[42,43,44]). Existing solutions and functionalities, such as described under state of the art, mainly remain limited to proof-of-concept demonstration. Examples of integration of such modular approaches and application to real-life challenges, such as our recently developed total analysis system for paper-based allergen detection (Fig. 8), remain rare. Therefore, applications that have arisen thus far from the field are limited. PHOTONFOOD will take up the challenge of taking several fundamental concepts that have been established by scientists at WU and develop those to reach a higher technology readiness level, with special focus on integration, (semi-) automation and user-friendliness. The foreseen outcome of the project will serve as a demonstrator to the paper microfluidics community on how this technology can be adapted to reach end-users, and to the food safety community on the advanced tools that are at our disposal for really getting analyses out of the lab and into the field.

More specifically, PHOTONFOOD will move beyond the state of the art by (i) the use of surface chemistry to provide selectivity in paper microfluidics, as a way to purify and enrich specific analytes, which has not been reported to date; (2) performing subsequent, or multi-stage liquid-liquid extractions in a continuous fashion (rather than a batch approach), which is possible due to the layered nature of 3D paper microfluidics; (3) using existing (immunochemistry) and innovative approaches to concentrate sample on specific regions of paper that can be coupled directly to MIR spectroscopic detection.

1.4.1.2 Tailored light sources

State of the art

In the past two decades, semiconductor lasers based on interband and quantum cascade laser technology have emerged as important light sources for miniaturized MIR analyser and sensing devices. Both technologies offer the advantage of providing high out-put powers tailorable to specific MIR spectral bands, small dimensions and long lifetimes. Since the experimental proof of QCLs in 1994[45] (1) and in 1997[46] (2) for ICLs respectively, a rapid technological development, in particular for QCLs, has resulted in a variety of commercially available

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QCLs and ICLs. Single mode distributed feedback (DFB) interband cascade lasers in the 1800-3500 cm^{-1}(3-5 µm) range were commercialized by NP starting from 2013.

To date, these single mode ICL and QCL based DFB lasers dominate the market with inherently very limited tuning ranges in the order of 10 cm^{-1} range. Recently, external cavity EC-QCLs with a wide tuning range have become commercially available for laboratory-based operation, covering up to 200 wavenumbers or more. In an EC-QCL the laser gain chip is part of an external cavity that can be mechanically tuned. The latter strongly hinders their application outside the controlled environment of the laboratory due to their lack of robustness, rather large size and exceptionally high costs.

Thus, the currently available technology for the MIR of mycotoxin detection is limited in twofold: The **tuning range is limited,** and the **cost of the technology is very high** and technically not rugged enough for out-of-lab applications. The PHOTONFOOD project requires two different kinds of light sources for the MIR, both suitable for on-site and outside-laboratory applications in the field, one source being cost-effective and easy to operate and another one that allows for high precision measurements. There are two promising technologies emerging to realize the two required light sources. Cost-effective, easy to operate incoherent light emitting diodes based on ICL material[46,4748,49] on one hand and high performance monolithically and widely tunable light sources based on the Vernier effect for the high-fidelity device of contamination detection on the other hand.

However, neither of these two emerging technologies has yet been pursued for the technologically much more challenging 1500-1800 cm^{-1} spectral region for which specific epitaxial structures based on a cascaded band diagram and specific chip technologies as well as the corresponding designs have to be developed to full-fill the requirements for the PHOTONFOOD project.

Up till now interband cascade light emitting diodes (IC-LEDs) have been realized by project partner NP in the 2500-3333 cm^{-1} wavelength region operating at room temperature with up to several mW of optical output power[50]. However, the devices have not yet been fully commercialized due to the challenges with respect to thermal management and flip chip bonding as an efficient device requires the out-coupling of the light through the semi-transparent substrate. Moreover, the spectral coverage is limited to a range of approximately 500 cm^{-1}. Many realizations of widely tunable single mode semiconductor lasers exist whose tuning principle is based on the Vernier effect. In order to be cost-effective, for industrial applications it is highly desirable that these semiconductor lasers are monolithic devices that are controlled only by the chip temperature and current levels. An intrinsic impediment in any Vernier-based tuning approach is the high dimensional parameter space and the relatively high sensitivity of the tuning of the laser to even small changes in temperature and current levels to enable cost-effective and practicable test procedures. This feature is achieved by novel design based on individually heated coupled cavities with an electric heater on heat spreader. These lasers possess a six-dimensional parameter space (chip temperature, two laser currents and three heater currents), but the tuning principle is simple enough, so a characterization of the devices is possible by only measuring a small number of tuning parameters. NP fabricated and characterized lasers based on this principle in the spectral region around 5000 cm^{-1} and achieved a tunability of the devices over a range of up to 500 cm^{-1} around a central emission wavelength of 4300 cm^{-1}.

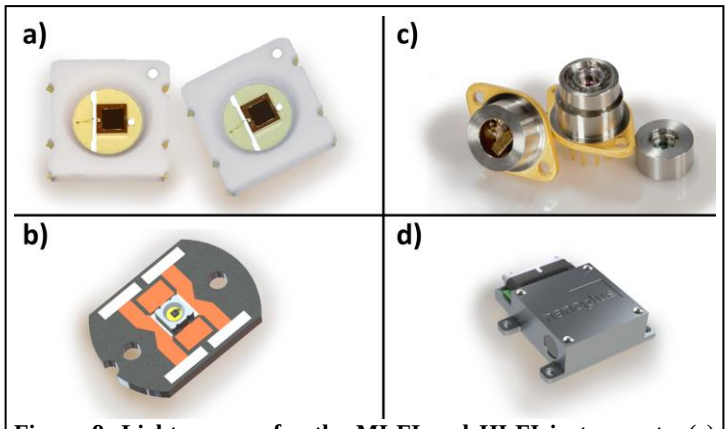

Figure 9: **Light sources for the MI-FI and HI-FI instruments.** (a) Photos of IC-LED chips mounted into SMD (surface mountable device) packages based on AlN for excellent heat removal. (b) CAD model of an AlN package with an IC-LED chip soldered onto a metal PCB, e.g. for integration into a measurement set-up or high-power optical characterization (c) TO3 housings for QCL chips with and without collimation lens. (d) The likely format of tunable laser package; 5-6 electrical drive currents might be used to achieve the required wavelength.

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PHOTONFOOD going beyond the state of the art

Within PHOTONFOOD we will realize two kinds of novel light sources that are tailor-made for the specific requirements of contamination detection in a Medium (MI-FI) and a High (HI-FI) fidelity device (see Fig. 9). For the MI-FI device we will realize the first long wavelength IC-LED for the 1500-1800 cm^{-1} spectral region operating at room temperature with optical output powers in the mW range. Due to the nature of the technology, MIR IC-LEDs are much more cost-effective than the fabrication of high precision single wavelength or tunable laser devices. Despite the chip technology itself we will realize a surface mountable device (SMD) as it is nowadays standard for LEDs in the visible to NIR spectral region in low-cost applications. The latter is quite challenging as the substrate material GaSb is thermomechanically more challenging with respect to CTE-matching to standard SMD packages used for LEDs such as Cu based lead frame packages integrated into QFNs (Quad-flat no-leads) or electro-plated AlN or Al_20_3 based ceramics with electrical vias. For the HI-FI device a widely tunable cascaded laser with emission wavelengths at approximately 1500-1800 cm^{-1} will be developed based on a specific laser device design based on the Vernier effect. Great care is taken to allow for a precise control of the parameter space required for laser operation to keep its operation simple and rugged. This feature is achieved by novel design based on individually heated coupled cavities with an electric heater on heat spreader approach. These lasers possess a six-dimensional parameter space (chip temperature, two laser currents and three heater currents), but the tuning principle is simple enough, so a characterization of the devices is possible by only measuring a small number of tuning parameters. The proposed widely tunable light source will provide essential advantages with respect to ruggedness, compactness, speed and costs compared to tunable external cavity QCL systems currently available on the market. NP's expertise in the design and realization of novel light sources based on interband cascade and QCL technology is vital to the successful realization of the project.

1.4.1.3 MIR sensing solution

State of the art

UULM along with partners NP, NMBU, BOKU, and IRIS have shown and continue to show in multiple EU-projects (MIRACLE, MYCOSPEC, TROPSENS, VOGAS, see Table 4) the utility and molecular specificity of MIR spectroscopy, sensing, and diagnostic concepts[28,51,52,53,54,55,56,57]. In particular, using tunable quantum cascade lasers (QCLs) enable recording high-fidelity/high-precision MIR spectra with sufficient signal strength and spectral resolution for identifying individual components even in complex matrices[58,59]. Next to the light source, a frequency-matched sampling interface is required for ensuring maximum sensitivity and reproducibility during the measurement. The most efficient, and least lossy approach to interact photons with sample molecules is by using an evanescent field generated at the interface between a high-refractive index waveguide and the usually lower refractive index sample, independent of its physical state (i.e., liquid or solid). UULM and partners have pioneered generic single-mode thin-film waveguide technologies that ideally combine with MIR laser light sources such as QCLs providing a homogeneous evanescent field for maximizing the interaction between MIR radiation and the sample molecules. They have already demonstrated their utility for mycotoxin analysis[22]. However, established semiconductor (i.e., GaAs/AlGaAs and HgCdTe) waveguides are of limited stability in real-world environments. Hence, in preparation of this project partner UULM has established the foundation of an entirely new generation of thin-film waveguides that fulfil the requirements of extreme robustness, resistance against chemical stress, and utmost mechanical stability, i.e., **thin-film**

[51] Sieger, Markus et al. (2018). Open Access Repositorium der Universität Ulm. http://dx.doi.org/10.18725/OPARU-6027
[52] G. Kos, et al. Food Additives & Contaminants: Part A, 2016, vol. 33(10), pp. 1596-1607,
[53] J. Haas, et al. Opt. Express 2019, vol. **27**, pp. 23059-66.
[54] López-Lorente, Á. I., et al. *Physica Status Solidi (A) Applications and Materials Science*, 2016, vol. *213*(8), pp. 2117-2123
[55] J. Haas, EV Catalán, P. Piron, M. Karlsson, B. Mizaikoff. Analyst. 2018, vol.143(21), pp. 5112-19.
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[57] J. Glöckler, et al. ACS Sensors, 2020, vol. 5, 4, pp. 1033–1039.
[58] Padilla-Jiménez, Amira C. *Vibrational spectroscopy studies of biomolecular systems: from amino acids to microorganisms*. Diss. 2014
[59] Zhang, Zhe. *Evaluation of infrared QCL, Synchrotron and bench-top sources for cell imaging in aqueous media*. Diss. The University of Manchester (United Kingdom), 2017.

waveguides made from nanocrystalline diamond (NCD)[55,56,60] . While being transparent in the MIR, such waveguides are chemical- and wear-resistant, and may be used for solid as well as liquid analysis[61,62] .

The proposed MI-FI device technology taking advantage of MIR IC-LED technology instead of high-end QCLs does not exist to date and is one of the pioneering innovations of this project, similar low-cost near-infrared (NIR) spectrometers (e.g., ConsumerPhysics SCIO) already exist on the market . However, the NIR region of the electromagnetic spectrum is not specific enough to detect contamination at levels that appear in food contamination. In addition, the spectral information produced by such devices has to be considered of very low fidelity[22]. We will indeed attempt developing the lowest-cost MIR analyzer on the market to date in the 2-4 k€ price segment. Given the significantly higher information content of MIR vs. NIR spectra (i.e., fundamental vibrations vs. overtone bands of liquid and solid samples), it is inherent to this spectral range that better and more information will be provided by any MIR device in contrast to its NIR siblings.

PHOTONFOOD going beyond the state of the art

PHOTONFOOD will go beyond the state-of-the-art in several ways. There is no high-precision portable, label-free, direct detection MIR device for chemical and microbial contamination analysis in food and food commodities currently available – neither at the research stage nor at the market – that allows generating high precision molecular signatures of microbial and chemical contaminants in the whole fingerprint MIR region (1800-900 cm^{-1}) without an expert user. Fig. 10a illustrates the key components of such a device that will be brought together for the first time by PHOTONFOOD in a suitcase-style portable dimension, i.e., (i) novel tunable EC-QCL, (ii) thin-film waveguide assembly with direct microfluidic paper-strip readout, and (iii) a single IR detector, whereby (i) and (ii) are innovative key developments of PHOTONFOOD. Fig. 10b illustrates the breakthrough concept of a low-cost, hand-held MIR IC-LED based analyzer combining as key components (i) an MIR IC-LED with (ii) a sampling interface accommodating the same microfluidic paper-strip readout as the analyzer shown in Fig. 10a, and (iii) an IR line detector array. Again, (i) and (ii) are innovative key developments of PHOTONFOOD, as is for the first packaging a MIR spectrometer into such a small and low-cost package. The combination of advanced MIR ICL/QCL technology with unique, exchangeable, on-chip microfabricated, thin-film NCD waveguides packaged into a user-friendly sampling interface will yield the first field-deployable HI-FI photonic devices ready for field testing along the entire food processing value chain and for all levels of stakeholders. As this technology will be augmented by low-cost paper-based sampling strategies, the cost per analysis will remain at a low and market-compatible level. While presenting a breakthrough technology, on-chip NCD waveguides have to date not been batch-fabrication at a wafer-scale, and have not been packaged into an end-user-friendly sampling interface such as the MIRa-Guide™ platform, as proposed within PHOTONFOOD. The waveguides have been commercialized by UULM and partners as an outcome of the MYCOSPEC project (see Table 4)[22]. However, for detection of contaminations in food matrixes, they need to be adapted to work in concert with low-cost, paper-based sampling strategies as a label-free direct photonic read-out tool (Fig. 10a).

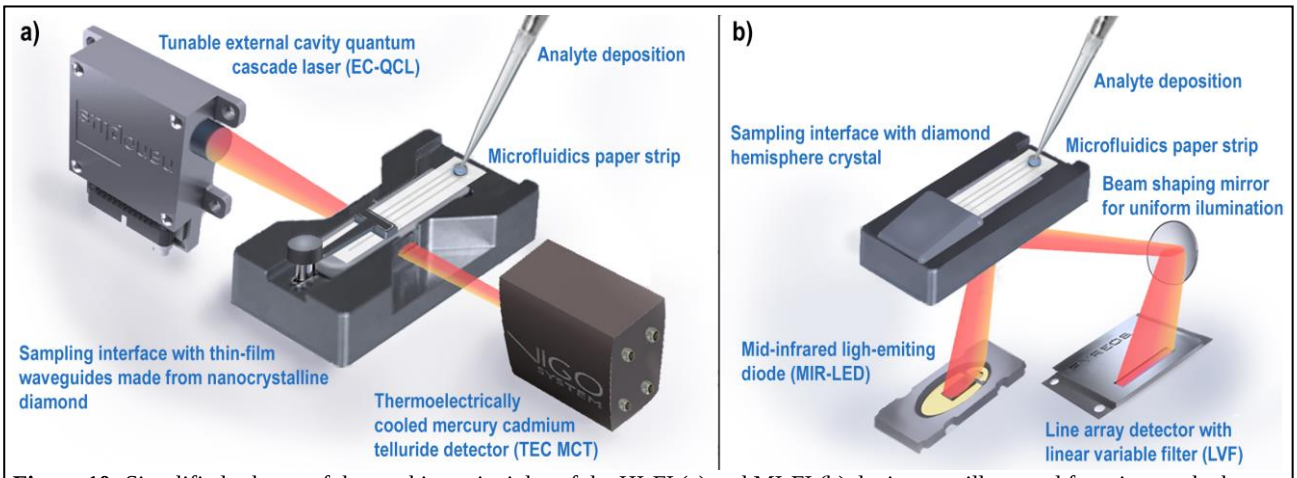

Figure 10: Simplified scheme of the working principles of the HI-FI (a) and MI-FI (b) devices are illustrated focusing on the key innovative components developed within PHOTONFOOD and omitting additional optics/electronics. The HI-FI device is based on an ICL/QCL laser, while the MI-FI device is based on low-cost MIR-LEDs.

[60] X. Wang, *Anal. Chem.* 2014, vol. 86(16), pp. 8136–8141.
[61] J. Haas, P. Artmann and B. Mizaikoff, ***RSC Adv.***, 2019, vol.9, pp. 8594-8599.
[62] X. Wang, Anal. Chem. 2013, vol. 85(22), pp. 10648–52.

To date, also no low-cost, monolithic, hand-held MIR IC-LED based MIR devices without any moving parts operating in the 1500-1800 cm^{-1} spectral window (i.e., the same spectral window as the HI-FI device) have been developed that allow collecting medium precision molecular signatures of fungal/mycotoxin contaminants. MIR radiation emitted from MIR-LEDs will be coupled to a single-bounce diamond hemisphere serving as a total internal reflection element producing a localized evanescent field, which constitutes the actual sample/photon interface (Fig. 10b). The essential sensing principle is equivalent to the HI-FI device, yet, using a single-bounce waveguide results in reduced sensitivity. The radiation will then be projected onto a line-array detector. It is anticipated that entire device will be largely manufactured from plastic to ensure low fabrication cost, yielding a MI-FI device targeting the 2-4 k€ price segment. The MI-FI device will interface with the same low-cost paper-based sampling strategies, thus even further reducing the cost per analysis.

1.4.1.4 Advanced data analysis for HI-FI and MI-FI solutions

State of the art

MIR spectra of multi-constituent samples are highly complex due the superposition of the spectral features in biological samples. Paper-based pre-concentration and separation will allow to reduce complexity considerably. Therefore, we expect that the commodity which is usually influencing a spectral signature in spectroscopy will be less interfering with the analyte signal because of the unique paper-based approach suggested by PHOTONFOOD. However, it is still expected that the obtained signal will contain interferent signals from the food matrix. A prerequisite for multivariate classification of spectral data is a separation of the corresponding chemical absorbance signals from unwanted interferences due to variations in sample thickness, noise, and scattering effects. Therefore, spectra need to be standardized, instrumental variability, atmospheric interference signals need to be removed and baseline variations need to be corrected. Therefore, the data pre-processing entails several essential steps: (i) estimation of the baseline; (ii) signal enhancement and noise suppression; and (iii) normalization[63]. Most spectral pre-processing procedures are based on using the whole spectral MIR range as it allows to estimate baselines and scaling effects in a reliable way.

Thermal broadband emitters require spacious components for wavelength selection which is in stark contrast to any size reduction approach. Hence miniaturized narrow emitters are suitable candidates to overcome this issue. In the MIRACLE project (see Table 4) NMBU is currently in collaboration with IRIS and UULM developing strategies for spectral standardization, when only a few (seven) wavenumbers are available. This has been solved by carefully selecting the seven single wavelength lasers in the MIR range to allow for a maximum of reliable standardization.

After the pre-processing step, MIR spectra need to be calibrated for chemical and microbial contamination levels. MIR spectroscopy has been used since the 90ies as a lab-based technique in combination with machine learning algorithms for the detection of microorganisms[64,65,66]. In combination with reference databases and powerful machine learning tools, microorganisms can be identified via their unique MIR molecular signature. The developed methods are in use as lab-based methods in clinical environments and for applications in the food sector, However, all established approaches are based on pure cultures of microorganisms, i.e. microorganisms need to be grown in pure cultures for 24-48 hours before they can be identified. No method exists for the detection of microorganisms or chemical contaminants that can be used on-site. Moving from state-of-the-art lab applications to field measurements in the food chain requires two main changes: (i) In order to be able to measure tiny amounts at ppm level, contaminant preconcentration and separation is required. (ii) for portable and handheld solutions, a narrow spectral range needs to be selected to realize a cost-effective and compact device for obtaining powerful light sources in the MIR fingerprint region. The project will have two spectral ranges available. The range 2500-3333 cm^{-1} is developed and the range 1500-1800 cm^{-1} will be developed in the PHOTONFOOD project. These spectral ranges will be used for establishing calibration models. NMBU has an extensive toolbox of calibration methods ranging from chemometric tools such as partial least squares regression, tree methods such as random forest to machine learning and deep learning tools. Deep Learning nets have been recently used for the spectroscopic data and showed to be powerful in discrimination[6768]. The use of augmentation combined with deep learning methods has shown excellent

[63] A.Kohler, et al. in Comprehensive Chemometrics: Chemical and Biochemical Data Analysis, 2009, pp.139-162.
[64] D.Naumann, in Int. Society for Optics and Photonics, 1985, pp.268-269.
[65] D.Naumann, D.Helm, H.Labischinski, Nature, 1991, vol. 351, pp.81-82.
[66] D.Helm, H.Labischinski, D.Naumann, J. of Microbiological Methods, 1991, vol. 14(2), pp.127-142.
[67] Acquarelli, Jacopo, et al. Analytica chimica acta, 2017, vol. 954, pp. 22-31.
[68] Zhang, Xiaolei, et al. Analytica chimica acta, 2019, vol. 1058, pp. 48-57.

results[69]. The use of data based on few wavelengths only, has also been shown to be promising[70]. The experience of MIRACLE project (see Table 4) supports these observations.

PHOTONFOOD going beyond the state of the art

The data analysis strategy of PHOTONFOOD is illustrated in Fig. 11. Laboratory based detection of mycotoxins and fungal contamination are based on broadband spectra with high quality and high spectral resolution. Pre-processing tools that deal with interferents in broadband spectra are readily available and developed by NMBU[71]. These methods have been further developed for dealing with a very low number of wavelengths (for example, seven wavelengths in the MIRACLE, see Table 4) The PHOTONFOOD solution will provide two spectral qualities (MI-FI and HI-FI) with a short spectra bandwidth in the range 1500-1800cm^{-1}. As opposed to a laboratory setting, the application of the ATR (MI-FI) and waveguide-based (HI-FI) MIR spectral signals will require new preprocessing tools. The application of extraction combined with paper-based microfluidics leads to interferent signals from solutions and substrates. In addition, light sources covering a narrow wavelength range (as opposed to full spectra covering the whole MIR region) will require adaption of existing preprocessing algorithms. NMBU will further develop and refine the current models focusing on signal enhancement and interferent removal via optimization of preprocessing parameters for PHOTONFOOD.

PHOTONFOOD will build classification and calibration models. In the classification model, standard thresholds for contaminants are used in the classifier training and the result of prediction will be contamination or no contamination in a sample. In calibration models, the exact level of a respective contaminant is estimated. Benchmark calibration and classification models for predicting levels of fungal and mycotoxin contamination will be established using benchmark instrumentation at NMBU and UULM. Multivariate methods based on latent variables such as partial least-squares regression and partial least-squares discriminant analysis (PLSR and PLS-DA) are powerful methods that allow interpretation of their decisions looking at molecular spectral signatures. Thus, PLSR and PLS-DA will help to understand the rationale of the model, i.e. if detection is done based on specific molecular spectral signatures from the contaminants or due to matrix changes. This is an important information to be obtained in order to decide about the generality of a developed calibration or classification model.

For final calibration and classification model building, state of the art techniques such as deep learning approaches and random forests will be applied. The size of the data sets for calibration and classification model establishment is in all practical situations limited. Therefore, PHOTONFOOD will use two innovative techniques for augmentation of the data sets for calibration and classification: (i) Data sets will be augmented by mixing samples with known contamination content as described in the WP6 in different concentrations in order to obtain a high number of samples for HI-FI and MI-FI measurements. The advantage is that the datasets can be augmented considerably without requiring additional reference analyses. (ii) Recent research performed by NMBU in the MIRACLE project (see Table 4), showed that augmentation of data sets by interferent effects could improve the performance of classifiers considerably. PHOTONFOOD will use this brand-new research to develop powerful and robust classifiers for detection of contaminants in all validation and demonstration cases in the project. The developed strategies for augmentation combined with deep learning will be available as a general strategy for adaption of the technology for new cases to which the PHOTONFOOD technology will be adapted. The developed calibration/classification models and preprocessing methods will be

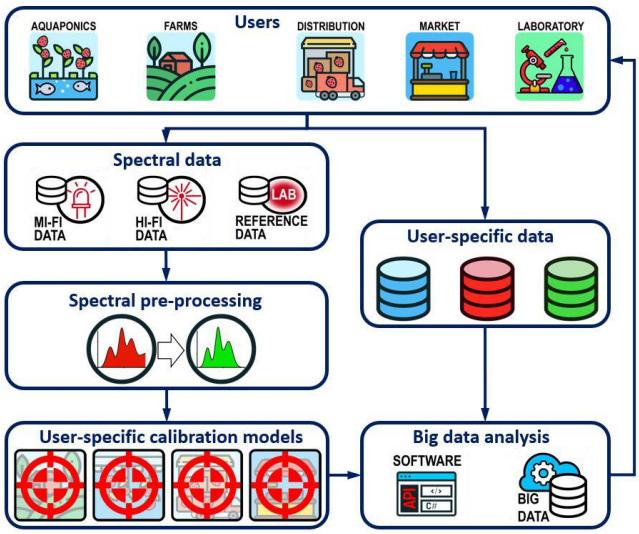

Figure 11: The data analysis consists of several important steps: (i) spectral pre-processing, (ii) calibration model building for contaminants and spoilers, (iii) and data integration for food safety improvement and waste reduction.

[69] EJ.Bjerrum, M.Glahder, and T.Skov, *arXiv preprint arXiv:1710.01927* (2017).
[70] R.Kasahara, et al. J. Biomedical optics express, 2018, vol. 9(1), pp.289-302.
[71] A.Kohler, et al. Comprehensive Chemometrics: Chemical and Biochemical Data Analysis, 2020, pp.83-100.

implemented in an easy-to- use interface software facilitating the use of it in a food supply chain. While PHOTONFOOD will develop specific models for the PHOTONFOOD cases, the developed strategies are easily transferrable for other types of microbial and chemical contamination.

A software platform will be designed and developed to analyse and readout the sensor measurements. In this way, the software platform will perform the digitalization of the data, integrating the models developed in the project, and providing a friendly user interface to facilitate the daily operations and decisions of the end-users involved. PHOTONFOOD will also assemble the results of detections in a mobile and web accessible interactive platform to provide multiple information to support decision-making in mycotoxin management by all actors in the food supply chain, including farmers, processors, manufacturers and retailers. A user-friendly decision support system (DSS) will be developed to reduce contamination in the food and feed supply chain, taking into account user requirements and the desired user experience of the target end-users (farmers, food and feed productions managers, among other stakeholders).

Our advisory partner IBM will, together with the consortium, develop a strategy for data integration along the food supply chain to analyze and utilize large amounts of data about food safety. The aim will be to detect critical points with respect to food safety and to improve food safety along the food supply chain (Fig. 7). The mobile and laptop application for contamination management to be designed and built in PHOTONFOOD, will take advantage of the software concept developed in the previous project MIRACLE (see Table 4), which main unit is composed of a local software application managing all the connected devices, and a local-cloud software to perform data synchronization with existing cloud/ERP infrastructures and allowing continuous updating and accessibility. It will be used an adapted version of its structured data of the API, based on unique URL, fast linking and readiness for future integration with ERP and business intelligence models.

1.4.1.5 Ambition in innovative case studies

By month 24 (milestone MS 4.1) the MI-FI and by month 30 (milestone MS 4.21) HI-FI devices will be validated at lab scale. KPIs are defined to assess the performance of the photonic devices in real settings, which will be done from month 36. The ultimate result of PHOTONFOOD are the final prototypes of the developed devices. We will provide detailed documentation concerning the development and the feedback from stakeholders. In order to define and calculate the KPIs and the suitability of the methods, we will take advantage of our end user and stakeholder involvement as these groups are to use the developed technology. The degree of accomplishment during demonstration will be described by utilizing adequate quantitative and qualitative indicators.

Case Study 1: Detection of contamination with Fusarium/Deoxynivalenol in wheat	
Lead partner	**Participant partners**
NEBIH	BOKU having own wheat fields as part of its Fusarium resistance breeding programme in cooperation with farmers from Europe
Motivation	
The occurrence of **fungal** and subsequently **mycotoxin** contamination in various crops is one of the major concerns in the field of cereals since it has significant implications for food and feed safety, food security and international trade. The EU produces yearly approximately 133 million tons (MT) wheat (~M€ 29038), 68 MT maize (~M€ 13571) and 8 MT oats (~M€ 1543). The annual loss due to mycotoxins is estimated at 5-10%, which equates to €1.2-2.4 billion (Bn) lost income for wheat alone. The reduction of these losses with only 1% would save €12-24 Mio. Grain and grain-based food products (e.g. pasta, bread, pastry) account for the largest contribution to mycotoxin exposure in all age classes of the EU population, in particular, due to the mycotoxins produced by *Fusarium* spp. and their most prevalent mycotoxin Deoxynivalenol (DON). **Pesticides**, in particular, insecticides, fungicides and herbicides are spread on wheat plantations. Residual of pesticides may remain being harmful for food safety. The control is based on food tolerances defined as the maximum residual limits (MRLs) established by European Union (EU), Codex Alimentarius (CA) and other organizations (EU Pesticides Database 2012). As a model pesticide we will use chlorpyrifos und chlorpyrifos-methyl which are widely used insecticides banned in the EU as of 2020.	
Challenges	

Despite huge research investments, the prevention and control of these toxic secondary metabolites have remained challenging. Further, contamination with pesticides residues increased last years due to the extensive use of different pesticides in wheat production. The agriculture and food industries are still vulnerable to problems of these contamination. While analytical techniques for mycotoxins and pesticides are improving and several commercial laboratories are available that provide screens for an array of mycotoxins and pesticides, the cost of analyses has been a constraint, in particular for SMEs. However, these costs may be insignificant compared with the economic consequences of production and health losses related to mycotoxin and pesticides residuals contamination. While newer immunoassays and wet chemistry technologies have reduced the cost of analyses, they frequently lack sensitivity in case of complex food matrices compared to laboratory chromatographic methods and may lead to false negatives. In addition, cross-reactivity of compounds can result in false positives. Comparatively long turnover times due to accurate, but laborious and expensive laboratory procedures decrease throughput and put additional financial burdens on producers demonstrating the safety of their products.

Technologies to be tested/validated

The PHOTONFOOD paper-based microfluidics, MI-FI and HI-FI devices and calibrations for fungi and DON mycotoxin contamination will be applied to measure *Fusarium* spp. fungi and DON mycotoxin contamination in wheat. The presence of fungi and DON will be detected by the MI-FI and HI-FI devices, which will be validated by microbiological culture methods for fungi and HPLC-UV/FLD and LC-MS/MS for DON mycotoxin.

In addition, HI-FI and MI-FI device and calibrations for pesticides will be applied for detecting presence o chlorpyrifos und chlorpyrifos-methyl. It is anticipated that the reference measurements via the HI-FI device will augment the low-resolution data provided by the MI-FI devices facilitating a multi-tier approach applicable to the entire food value chain.

Innovative approach, objectives, and contributions to the project

As an innovative approach, the novel handheld photonic MI-FI device will provide a cheap tool for monitoring fungal and mycotoxin contamination and detection of pesticides residues on a farm level. The objective of this case study is to demonstrate the applicability of the MI-Fi device for daily screening-based monitoring and the HI-FI device for on-farm reference analysis as tools for grain producers to prevent contaminated batches to be supplied into the food chain by implementing early monitoring and detection.

Using the PHOTONFOOD solution, a big land area can be sectioned, and sub-batches can be monitored separately for mycotoxin contamination and presence of pesticides residues. The presence of fungi and mycotoxin contamination is extremely heterogeneous in grain production. Therefore, special attention will be given to define the size of sub-batches and the elaboration of a reliable sectioning between them during the application of a handheld photonic MI-FI device. It is important to validated that the technology is prepared to be operated under extreme weather conditions, especially heat, dust and humidity. When routine monitoring with the MI-FI device detects fungal contamination, precise measurements with the HI-FI device can be performed on the same sample as reference analysis. By eliminating the problematic sub-batches, the contamination of the whole batch can be prevented.

Sustainability and applicability in other areas

The amount of food waste could be effectively reduced throughout the whole food chain, the income loss of the farmers and food processors could be minimized, and the consumers could enjoy a higher level of food safety. Cost effectiveness will be high. The MI-FI device for monitoring will be held under a price of €2,500- €3,000, the HI-FI device will be designed under €25,000- €30,000 as a market price (this is a highly competitive price, which is cheaper than a cheap FT-IR option). As will be seen, a significant means of keeping the costs of the technology affordable and competitively priced will be derived from the novel photonics technology that will be used in the system design, which can be batch-fabricated on a wafer scale, thus providing the potential for mass fabrication at reasonable cost. As more analysers are produced and more such devices will be needed significant price drops are expected.

Key performance indicators (KPIs) for the case study

- 100 samples of wheat will be measured for detecting Fusarium contamination by HI-FI and 500 samples by MI-FI;
- 100 samples wheat will be measured for detecting DON contamination by HI-FI and 500 samples by MI-FI;
- Fusarium contamination will be detected at the detection limit of 10 colony forming units per g (CFU/g) by HI-FI and 100 CFU/g by MI-FI;
- DON contamination will be detected within the detection limit of 1000 ppb by HI-FI;
- DON contamination will be detected within the detection limit of 2000 ppb by MI-FI;
- Pesticides residues of chlorpyrifos und chlorpyrifos-methyl will be detected in 100 wheat samples at the relevant maximum residual limits (MRLs) by HI-FI and MI-FI;

Case Study 2: Contamination of nuts with mycotoxins at distributer/EU entry point	
Lead partner	**Participant partners**
BOKU	NEBIH, Seeberger, BAMA and peanut producers from China Hualong Foodstuffs Enterprise facilitated through the MyToolBox EU-China-Mycotoxin Forum and ROMER, a supplier of official control laboratories.
Motivation	
Peanuts are one of the most commonly consumed snacks in the European households: In Europe, about 550k tons of peanuts (groundnuts) are consumed each year. The Netherlands, Germany, the United Kingdom, Spain, Poland and Italy have a significant importing and consuming role in Europe. The European Food Safety Authority has reported that peanuts are one of the commodities that are mostly jeopardized by mycotoxin contamination. The occurrence of mycotoxins in the peanut supply chain and the occurrence of in particular aflatoxin B1 (AFB1) in the case of other nuts, such as pistachio, is a significant problem. Notifications of over 400 border rejections by EU´s Rapid Alert System of Food and Feed (RASFF) have been reported since 2011 (RASFF database, accessed Jan 6 2019), thus the border inspection points and laboratories of the authorities should be reinforced by a mycotoxin monitoring device.	
Challenges	
An EC DG Health report (DG SANTE, 2017-6079) recommends that official controls of peanuts have to be carried out at all stages of production, processing and export. However, the analytical methods available today are tedious and the turnaround time for mycotoxin analysis is usually from 3 to 5 days.	
Technologies to be tested/validated	
The PHOTONFOOD paper-based microfluidics, the MI-FI device for screening and the HI-Fi device for reference analysis will be tested at the EU entry point (Seeberger), at wholesale (BAMA) and at a retailer (through BAMA). The sample preparation of oily seeds and nuts will be carefully planned and a reliable protocol will be elaborated. The MI-FI device will be used to screen a large number of batches and products over a period of 4 weeks. The developed paper-based microfluidics will be tested by the target users. Moreover, MI-FI and HI-FI devices will be validated by comparing the performance characteristics of the innovative photonic device with the reference methods (based on LC-MS/MS and HPLC-UV/FLD).	
Innovative approach, objectives, and contributions to the project	
PHOTONFOOD will allow the food chain actors, including authorities to carry out cost-effective and rapid testing, which is particular of relevance for heterogeneous aflatoxin contamination as encountered in peanut batches.	
Sustainability, replicability and potential use in other areas	
The experience of the peanut case study can be generalized to similar products such as almond, pistachio, hazelnut, walnut etc. The photonic-based solution can also be explored and applied in the future within EU-China-cooperations to control and detect aflatoxin levels in the peanut-supply chain as demanded by DG Health recently. The developments PHOTONFOOD can also serve as the basis for the advancement of the real-time post-harvest environmental monitoring system for storage of peanuts and especially in China.	
Key performance indicators (KPIs) for the case study	

- 100 samples of peanuts and any other types of nuts products will be measured for detecting *Aspergillus* contamination by HI-FI and 500 samples f MI-FI;
- 100 samples of peanuts and any other type of nuts products will be measured for detecting AFB1 contamination by HI-FI and 500 samples by MI-FI;
- Aspergillus contamination will be detected at the detection limit of 10 colony forming units per g (CFU/g) by HI-FI and 100 CFU/g by MI-FI;
- AFB1 contamination will be detected within the detection limit of 2 ppb by HI-FI;
- AFB1 contamination will be detected within the detection limit of 82000 ppb by MI-FI;

Case Study 3: Aquaponics – Presence of fungi monitoring in herbs production	
Lead partner	**Participant partners**
BOKU	BOKU, BIGH
Motivation	
Aquaponics is a sustainable method for producing fish and vegetables simultaneously, as it requires less water, land and labour than traditional agriculture. The introduction of these kind of innovative production systems (e.g. aquaponics) is fostered by the European Green Deal, societal expectations and consumer needs. Although the idea of aquaponics goes back to ancient times, it became an innovative production technology during the last decades, used by commercial farms as well. The market value of aquaponics is constantly increasing year by year. As it is becoming a widely used production technology, specialized controlling methods are needed to maintain food safety and quality at the farms' level.	
Challenges	
Particular challenges include the control of fungal (Oomycetes) contamination in the production of herbs and tomatoes and controlled usage of antibiotics and pesticides. During this case study, special aspects of aquaponics will be examined, providing safe, reliable and affordable monitoring solution for producers. As aquaponics is a relatively new and a constantly evolving agricultural system, producers have reported some typical challenges, such as the appearance of different types of moulds in the plant culture, including some mycotoxin-producing species. The early detection of fungal contamination is also challenging, especially for organic producers, such as our partner, BIGH. In addition, we will monitor antibiotics residues by using a model antibiotic polymyxin. Europe's animals consume more antibiotics than humans on average, often via livestock feeds on factory farms, where farmers routinely use them as a prophylactic against the occurrence or spread of disease. At least 25,000 people are dying annually in Europe from antimicrobial-resistant infections, and scientists have warned that without reform, routine medical interventions could soon become impossible. The new legislation, which will become law by 2022, bans the use of human reserve antibiotics in veterinary medicine and the use of unprescribed animal antimicrobials.	
Technologies to be tested/validated	
During this case study, the presence of fungi and mycotoxin will be measured by the MI-FI device combined with the paper-based microfluidics on a large number of samples in routine monitoring and paper-based microfluidics in combination with HI-FI devices for reference analysis. The accuracy of the two devices will be validated by reference methods: microbiological culture methods for fungi; HPLC-UV/FLD, LC-MS/MS for antibiotic and pesticide residues. In addition to aquaponics production, BIGH provides the opportunity to test the technology in on site processing and in vending on local supermarkets and grocery stores.	
Innovative approach, objectives, and contributions to the project	
Consequently, PHOTONFOOD will build on and further develop a MIR based testing platform, which will allow the food chain actors, including authorities to carry out cost-effective and rapid testing, which is particular of relevance for fungal, pesticides and antibiotics contamination encountered in aquaponic vegetable/herbs production peanut batches.	
Sustainability, replicability and potential use in other areas	
The results of this case study will be relevant for other innovative, complex production technologies, especially in urban agriculture, which is an important element of smart city programs.	
Key performance indicators (KPIs) for the case study	

- 100 pots of tomato and herbs and 100 water samples will be examined for detecting fungal Oomycetes contamination by HI-F and MI-FI;
- Oomycetes contamination will be detected at the detection limit of 10 colony forming units per g (CFU/g) by HI-FI and 100 CFU/g by MI-FI;
- 100 water samples from fish tanks will be examined for detection of polymyxin by HI-FI and MI-FI;
- 100 water samples from the hydroponics part will be examined for the presence of pesticides chlorpyrifos und chlorpyrifos-methyl by HI-FI and MI-FI;

1.4.1.6 Innovation potential

The PHOTONFOOD project will develop six innovations:

1) PHOTONFOOD will develop **3D paper microfluidics for purifications, enrichment, multi-stage liquid-liquid extractions** of specific analytes for MIR spectroscopy. Such technology does not exist today and will enable the use of spectroscopic techniques for detection and quantification of analytes at concentration that have been far outside reach until now.
2) PHOTONFOOD will develop **widely tuneable (multi-wavelengths), monolithic single mode semiconductor lasers in the 1500-1800 cm^{-1} spectral range** that are highly suitable for MIR spectroscopy of biomaterials. Currently only single-mode, one-wavelength DFBs based on QCL or ICL Technology are available (QCL: https://www.alpeslasers.ch/, ICL: https://www.nanoplus.com).
3) PHOTONFOOD will develop **LEDs with high output-power and low power consumption devices with a broad spectral range** as low-cost MIR sources. Currently there are no LED-like products are for this wavelength range on the market.
4) PHOTONFOOD will develop **portable and field deployable handheld photonic devices including bright laser sources tunable via novel concepts, advanced optical transducers uniquely compatiblae with paper-based microfluidic strips, and the first MIR-LEDs**. Currently available products with overlapping capabilities include ChemDetect™ Analyzer, (https://www.daylightsolutions.com/product/chemdetect/; weaknesses: no sampling interface provided, has to be developed by end-user; just light-source and detector packaged into an instrument); Agilent 4300 Handheld FTIR, https://www.agilent.com/en/products/ftir/ftir-compact-portable-systems/4300-handheld-ftir (weaknesses: has moving parts, reduced SNR, cannot be used with thin-film waveguides); MIRa-GuideR (https://www.mgopticalsolutions.com/produkte/mira-guide-atr-wellenleiter/; developed by UULM and MGOS in EU-project MYCOSPEC; will serve as a base component in PHOTONFOOD for HI-FI analyser; weaknesses: no thin-film NCD waveguides integrated, no paper-based microfluidic strip sampling interface), IRsweep IRis-F1 (https://irsweep.com/products/iris-f1/?gclid=EAIaIQobChMIuKuS4p_q6QIVieFRCh02rg1_EAAYAyAAEgLQrvD_BwE; weaknesses: cannot be used portable or in the field; no sampling interface provided)
5) PHOTONFOOD will develop a **MIFI MIR device with superior spectral quality addressing the MIR range needed for chemical analysis of food contaminants**. Currently there is only a comparable device available for the near-infrared region with low spectral quality and which does not cover the MIR range. The price level is comparable to the MI-FI device envisioned by PHOTONFOOD.
6) The PHOTONFOOD **software application will integrate specific calibration models and comprise intuitive decision support tools and friendly user-interface to facilitate user acceptance and on-site tests which are ready for future integration with ERP and business intelligence models**. There are currently no software applications for MI-FI and HI-FI contamination detection on the market with specific calibration models.

Prior to the project planning, a patent search was performed using WIPO PATENTSCOPE. The closest prior art identified included EP3634631A1 (pending), US8921118B2 and CN102016595B for the 3D paper microfluidics; and US8320728B2, US20170163001A1 (pending) and US20180329269A1 (pending) for the semiconductor laser. In both cases, the technologies developed by PHOTONFOOD may constitute novel, patentable subject matter and can therefore be regarded as breakthrough innovations on the technical side. The other PHOTONFOOD developments constitute product innovations. They will require the purchase of components and/or licensing of technology and may not necessarily be patentable in their own right. Due to exceptionally low price position, the system solution (portable and field-deployable handheld photonic devices…) can be regarded as a breakthrough innovation on the product development side.

2. Impact

2.1 Expected impacts

The present **market for fungal, mycotoxin, pesticides and antibiotics testing** (kits/commercial lab services) already has a global value of € 1 billion (where Europe is the largest geographical segment with approximately 40%). The ten globally leading providers include ROMER (partner in PHOTONFOOD), ALS, AsureQuality, Bureau Veritas, Eurofins, Intertek, Mérieux, Microbac, Neogen, Omic, SGS and Symbio. Although the market has been growing steadily at around 6.5% CAGR, there are **significant barriers to further development**:

- **Investment requirements on the customer side are still too high**: according to market research conducted by the consortium members, customer willingness-to-pay for instrumentation (ELISA readers and related equipment) tops out at around €3000 per lab bench.
- **Per-sample costs must be brought down:** according to market research conducted by the consortium members, customer willingness-to-pay for mycotoxin rapid tests (strip tests) tops out at €5-8.
- **Current staff skill requirements are unrealistic:** current analytical procedures require significant amounts of manual sample preparation and analysis by highly trained laboratory staff. The costs related to this are orders of magnitude higher than the costs related to instrumentation, reagents and test kits.
- **Hazardous waste needs to be minimised:** current analytical procedures are sample-destructive and generate significant amounts of hazardous waste that needs to be disposed of at high cost.

This has unfortunate direct consequences: **current analytical procedures are only affordable for food chain actors who operate at relatively high scale**. Small and medium-sized operators—and this applies to primary producers, importers, food manufacturers, retailers and food authorities alike—are forced to test as rarely as possible and at relatively large batch sizes. Whenever a test yields a positive result, **the amount of goods that needs to be destroyed leads to large losses**, reducing the competitiveness of small and medium-sized operators. This, in turn, has large consequences on the societal level: the level of food quality and safety in the affected value chains is not as high as it could be. Growth and employment in the involved business sectors are kept below the levels they could reach. Sustainable development goals become more difficult to reach.

The aim of PHOTONFOOD is to **bring down the cost of fungal, mycotoxin, pesticides and antibiotics testing** on all levels of the food chain by developing, validating and demonstrating a compact, portable, broadly adaptable, multipurpose solution that is photonics-based and therefore non-destructive by nature. Our solution will be designed to deliver the following benefits:

- **Lower investments**
- **Lower cost per sample**
- **Rapid results**
- **Higher throughput**
- **No special education needed for staff**
- **Training is simple**
- **Can be installed in various industrial environments**
- **Insignificant hazardous waste production**

The **direct and indirect impacts** of our solution, with respect to relevant **EU policy objectives** in the areas of food quality and safety, reduction of food waste, strengthening of small and medium-scale farming and local or novel ways of food production and processing, reinforcing value chains and increasing the competitiveness of the European photonics sector, and access to risk finance are detailed in Table 7. In addition, we detail the impacts with respect to the **Photonics Strategic Roadmap** and the UN **Sustainable Development Goals**. For each expected impact (EI), we specify which strategic objectives (SO) the project has adopted in order to ensure that the impact will be delivered, and KPIs and target levels that will allow the verification of whether the impacts have been achieved in the medium term (within seven years from project start) and long term (within 9 years from project start).

Table 7. Expected impacts (EI) connected to the specific objectives (SOs) and key performance indicators (KPIs)

Expected impacts related to the ICT-37-2020 call
Cost-effective, flexible and easy-to-use photonic-based quality control analysis / SO1 – SO9

The PHOTONFOOD solution will reduce the amount of costly, specialized, time-consuming laboratory work by providing a validated portable photonic solutions that can be operated effortlessly on farms, in commodity distribution, food inspection and border control, food processing and manufacturing, and retail. **KPI Decrease in analysis cost:** at least 50% decrease in per-sample analysis cost within 7 years from project start (target level for prototype stage: < €3.00 per sample, target level after full scale-up: < €0.75 per sample for the MI-FI solution). **KPI Reduced instrumentation cost:** 50% decrease in instrumentation cost within 7 years from project start (target level after full scale-up: market price < €2,500 per MI-FI solution and €25,000 per HI-FI solution). **KPI High user acceptance:** The usability of the solution and its acceptance among users and stakeholders will be verified during the lifetime of the project. 90% of the users will be satisfied with the project results, and 85% will be willing to use it for quality control.
Significant reduction of food waste in the farm-to-fork food production chain / SO1 – SO9
The PHOTONFOOD solution will enable faster verification of suspected contamination cases at smaller batch sizes, increasing the ability to localise the contamination and thereby reducing the number and size of batches that have to be destroyed upon suspicion alone, reducing the amount of food waste in the relevant food value chains. **KPI Reduction of food waste:** 20% decrease in food waste streams in relevant European value chains (cereals, nuts, dry fruits and aquaponics industry) within 9 years from project start.
Strengthening small/medium-scale farming / SO1 – SO9
Key parts of the modular solutions developed and validated by PHOTONFOOD will be fully usable without a need of further laboratory analysis, making it usable and affordable for small and medium-sized primary producers. This will enable small and medium-sized producers to increase the quality and safety of their products and thereby attain a better competitive position relative to large producers having sufficient resources already now to ensure the quality and safety of their production. **KPI Increased quality of food:** Increased quality of farm-produced food by 50% within 10 years from project start.
Strengthening local or novel ways of food production and processing / SO1 – SO9
The PHOTONFOOD solution will be developed with aquaponics and vertical farming as specific production test cases and local farmer's markets as a specific retail test case. This will ensure that the equipment is fully adapted to novel and local production systems, ensuring that they can achieve the same level of quality and safety as already established production systems. **KPI Increased contaminants monitoring in novel production:** Penetration of 20% (turnover-based) of the EU aquaponics sector within 7 years from project start; Penetration of 40% (turnover-based) of the EU aquaponics sector within 9 years from project start.
Expected Impacts related to the Photonic Strategic Roadmap[72]
Reinforced value chains and deployment of photonics technologies by stronger cooperation of photonics stakeholders, clusters and end-users / SO1 – SO11
The PHOTONFOOD consortium is a multi-actor initiative bringing together researchers (in the fields of photonics, data science, fungal biology, analytical chemistry and food safety) with commercial testing providers (such as ROMER), photonics SMEs (such as NP), end-users in primary food production (such as BIGH), processing and wholesale (such as BAMA and Seeberger) and food control (such as NEBIH). The project will serve as a model for future collaboration, laying the foundations for a strong photonics value chain and its future development. **KPI Increase in photonics-based food analysis:** Increase in the global market share of photonics-based mycotoxin testing to 20% within 7 years from project start. Increase in the global market share of photonics-based mycotoxin testing to 40% within 10 years from project start. **KPI Increase in photonics-related projects and developments:** The number of collaborative development projects in the photonics value chain will substantially increase from the start and after the project.
Increased competitiveness of the European photonics sector / SO1 – SO11
The validated PHOTONFOOD solution will sharply increase the number of realistic use cases on all levels of the food chain and in related industries. At the same time, the tiered precision levels (high-, mid- and low-fidelity) will considerably reduce the costs of photonics-based detection solutions. Together, these

72 https://www.photonics21.org/

effects will substantially boost the competitiveness of European photonics sector vs. American and Asian competitors, and also vs. technological substitutes in the domain of traditional laboratory analytics. **KPI Increase in global market share:** Increase in the global market share of European-produced (or out-licensed) photonics-based fungal/mycotoxin testing equipment to 10% within 7 years from project start. Increase in the global market share of European-produced (or out-licensed) photonics-based fungal/mycotoxin testing equipment to 20% within 9 years from project start. **KPI Increase in number of users:** The number of customers using photonics-based contamination detection solutions will increase in all relevant customer segments (primary producers, processor and wholesale, retail, food control authorities) 30% after 7 years from the project start.
Improved access to risk finance for the photonics sector in Europe / SO1 – SO11
Increases in R&D and business development dynamics in the European photonics sector will make it an attractive opportunity for venture capital funds and private equity investors. **KPI Increase in investments:** Increase by €25 mio in Series A investments in European companies developing photonics-based fungal/mycotoxin testing equipment within 7 years from project start. Increase by €75 mio in Series B investments in European companies developing photonics-based fungal/mycotoxin testing equipment within 9 years from project start
Expected Impacts related to the Sustainable Developments Goals (SDGs)[73]
Increase in healthy non-contaminated food to decrease in mycotoxin-related health burden under "SDG3 Ensure healthy lives and promote well-being for all at all ages" / SO1 – SO9
The deployment of the PHOTONFOOD technology in major European food chains will contribute to the general improvement of food quality and safety, reducing the health burden arising from mycotoxin contamination of cereals, nuts, dry fruits and aquaponics products. **KPI Increase in food quality:** 10%-point decrease in the positive sample rate in member state mycotoxin surveillance programmes within 9 years from project start.
Promote integration of innovative sensor solutions in food and farm industry under "SDG9 Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation" / SO1 – SO11
PHOTONFOOD will develop highly flexible innovative sensors for the analysis of microbiological and chemical contamination which can be easily deployed in the production facilities of farms, food processing and aquaponics facilities. **KPI Increase in integration of photonics analysers in food safety control:** 40% increase in the sales of photonics equipment for process control in the cereals, nuts, dry fruits and aquaponics value chains within 9 years from project start.
Decrease of the food waste as an urgent action to combat climate change under "SDG13 Take urgent action to combat climate change and its impacts" (long-term EI) / SO1 – SO9
The deployment of the PHOTONFOOD technology in major food value chains will allow the detection of microbial and chemical contamination at very early stages and enable early countermeasures, significantly reducing the amount of food spoilage. **KPI Decrease in food waste:** 20% decrease in relevant food waste streams (cereals, nuts, dry fruits and aquaponics industry) within 9 years from project start
Other Expected impacts
Increase food yield, quality and safety / SO1 – SO9
PHOTONFOOD will develop a flexible photonic sensing solution field-tested in case studies allowing rapid detection of fungal and mycotoxin contamination on all levels of the food chain (production, processing, import, wholesale, retail, food control), increasing the quality and safety of the European food supply. **KPI Increase in sustainability of food production:** 10%-point decrease in the positive sample rate in member state mycotoxin surveillance programmes within 9 years from project start.
Ensuring high food quality standards at lower resource usage and competitive costs / SO1 – SO9
The deployment of the photonics-based technology developed in PHOTONFOOD will reduce the need for costly, specialized, time-consuming laboratory analytics. Furthermore, the modular concept of the platform ensures that the precision of contamination detection can easily be adapted to desired (and affordable) target levels.

73 https://sustainabledevelopment.un.org/?menu=1300

KPI Decrease in analysis cost: at least 50% decrease in fungal/mycotoxin, pesticides and antibiotics per-sample analysis cost within 7 years from project start (target level for prototype stage: < €3.00 per sample, target level after full scale-up: < €0.75 per sample for the MI-FI solution). **KPI Increased availability of photonics analysers for food analysis:** 50% decrease in instrumentation cost within 7 years from project start (target level after full scale-up: market price < €2,500 per lab bench for the MIFI solution).
Societal impact on the increased quality and robustness of public food control services / SO1 – SO9
PHOTONFOOD will develop technology which will allow public food safety and border control authorities to perform more tests faster but at the same overall cost as before. **KPI Increase in the robustness of food safety monitoring:** 50% increase in the number of fungal/mycotoxin samples analysed by European food safety authorities within 9 years from project start.
Economic impact on the growth of photonic industry and photonics-related jobs / SO1 – SO11
Broad deployment of the technology developed by PHOTONFOOD for food chains, and of further innovations based on the PHOTONFOOD approach but targeting other value chains, will significantly contribute to the growth of the European photonics industry in general. **KPI Increased growth of innovative photonics industry:** 30% increase in the (constant-price) sales of European photonics companies and 10% increase in the total number of photonics-related jobs in Europe within 9 years from project start.

2.2 Measures to maximise impact

Measures to maximise impact will be delivered via WP7 Communication, Dissemination and Exploitation, led by accelCH and closely supported by NEBIH, NP and ROMER. Support and feedback on the planned measures and their long-term impact for commercial and non-commercial exploitation, will be received from a dedicated Innovation Manager. The primary goal of our multi-disciplinary and cross-sectorial consortium in WP7 is to disseminate the research results generated in the scientific work packages (WP1-WP6), to raise awareness and provide information on our novel photon-based technology, through communication and outreach activities, and to pave the way for exploitation of the PHOTONFOOD innovations and IP for medium and long-term non-commercial and commercial purposes such as detection and analysis of (micro)biological and chemical contamination along the food production chain from 'farm to fork'. An overview of our dissemination, exploitation, and communication activities is detailed in the following sections (see 2.2.1 and 2.2.2).

2.2.1 Dissemination and exploitation of results

To maximise impact, we have defined a project specific dissemination, exploitation and communication strategy that builds on the following three main points.

(1) Stakeholder involvement and target-group orientation: To ensure that the interests and requirements of external audiences are fully taken into consideration during the delivery of the PHOTONFOOD project. Along the whole food value chain 'from farm to fork', we will actively involve representatives of our key target audiences (farmers and novel food producers, food processors, food retailers and whole sellers, analytical labs, EU entry points, technology providers, photonic companies), from the beginning of the project to ensure technology acceptance (WP1), validation in a realistic setting (WP6) and outreach (WP7). The proposed target-group orientation allows us to define and implement measures that are geared to the specific needs of the different audiences (Fig. 12). Additionally, PHOTONFOOD will be in close contact with relevant food safety authorities and will work towards strengthening the community, uniting many important players in food supply chain. The project

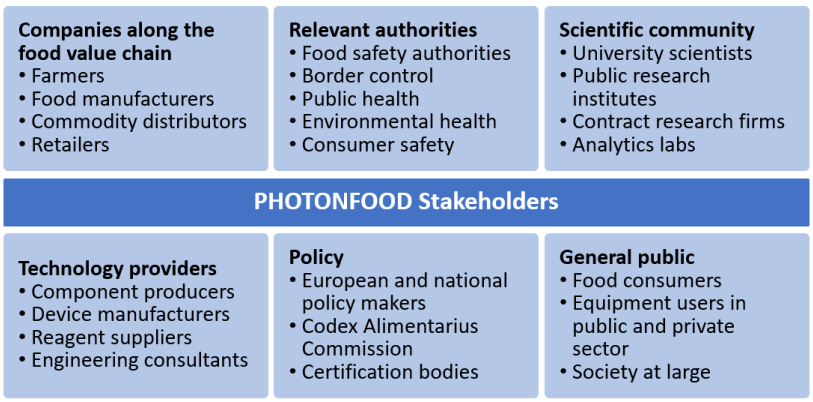

Figure 12: PHOTONFOOD's key stakeholder groups

will aim to collect feedback from interested and influential stakeholders, as well as related projects, and disseminate its results and achievements to them, in order to maximise impact.

(2) Cross-channel and multi-media approach: To guarantee the widest possible outreach, we will use different channels to inform about the PHOTONFOOD project and to disseminate research results and innovation and technology developments. These range from direct or face-to-face contact with scientists at conferences and webinars and industrial workshops organized in WP1 to indirect information distributed to all stakeholders. Additionally, we will take advantage of print media, digital tools and platforms for our dissemination, training, and communication activities. Some of our digital outreach highlights include video clips, open webinars, education and marketing videos.

(3) Open access and open data policy: To enable the future uptake of PHOTONFOOD's research results and to facilitate future sustainability of the materials and technologies, we are fully committed to open access publishing, following the Guidelines on Open Access to Scientific Publications and Research Data in Horizon 2020 that are in line with current trends in IP management and publication strategies (see section 2.2.1.3). This will be achieved by (i) applying existing standards for the development and results publication, (ii) involving stakeholders as of the beginning of the action and 3) publishing project results in peer-reviewed journals while ensuring open access.

Plan for dissemination and exploitation of results

The WP7 leader accelCH will coordinate and monitor the related tasks and closely cooperate with all project partners and external collaborators involved. They will be responsible for monitoring and coordination of the implementation of the defined outreach strategy, through the dissemination activities, exploitation plans, and communication actions at the consortium level with all work package leaders offering support related to dissemination, communication and exploitation of the project and its outcomes. To implement the Dissemination and Exploitation strategies, we will set up two separate plans for the dissemination and exploitation of the project's results:

Table 8: PHOTONFOOD plans to implement the communication, dissemination, and exploitation strategy

Communication and Dissemination Plan (CDP)	**Roadmap for Exploitation** (RfE)
Content: The CDP will include a refined strategy, planned activities as outlined in WP7, and described in more detail below, their expected impacts. The CDP will further include a detailed stakeholder analysis and a definition of the most relevant tools and channels to address the individual PHOTONFOOD target audiences. **Objectives**: The CDP will define the strategy and implementation of all PHOTONFOOD dissemination and communication activities and be continually assessed and updated, to achieve the widest possible outreach to our defined target groups. **Lead:** Guided by accelCH with many years of experience as Dissemination and Exploitation Manager in many EU funded projects, all partners will contribute necessary information and input to the CDP.	**Content**: Comprising a plan for the future commercial and non-commercial use of results at Consortium level, group, and individual partner level. The RfE will cover the exploitation, starting during the lifetime of the project (short-term) and mid and longer-term plans <4 years, and <10 years after the project ends). **Objectives**: The RfE will build upon the initial technology and economic evaluations and requirements from WP1, as well as outcomes from WP6, to clearly define and develop possible business and operational models for the project results. **Lead**: The RfE development will be led by HS.
Deliverable D7.2, M9, Task 7.1	Deliverable D7.6, M48, Task 7.4

Assessment of outreach measures. To assess both the implementation and impact of the defined activities, we will include evaluation of our individual outreach activities, based on the guidelines and structure set out in the European Commission's Toolkit for the evaluation of the communication activities for small scale evaluations[74] This will include defining the scope, objectives, and metrics for each outreach activity defined in the CDP, as well as quarterly monitoring and updates of the activities to see if interventions are required (D7.3 Interim communication and dissemination activities report, M24). This allows for assessment of the individual activities and timely adaption, where relevant, to maximise outreach.

Dissemination Measures. Publicly disclosing PHOTONFOOD results and outcomes to targeted stakeholders, sharing our knowledge to facilitate discussions on food safety and monitoring of relevant contaminants, novel

[74] European Commission, July 2015. *Toolkit for the evaluation of the communication activities*, Brussels: 6.

detection technologies and food safety digitalisation, circulating scientific validation results to the scientific community, and informing policymakers on PHOTONFOOD are the main aims of our dissemination plan. In order to achieve this, we have planned a broad range of cross-channel and multi-media dissemination activities, as shown in table 9.

Table 9: PHOTONFOOD dissemination activities

Activity and Description	**Target group | Lead**	**Impact | KPIs**
Conference presentations, peer reviewed publications Comply with Guidelines on Open Access to Scientific Publications and Research Data in H2020. Results to be published in peer-reviewed journals, e.g. Jounal of Biophotonics, Applied Spectroscopy, Food Control, Food Additives and Contaminants, Analyst, Analytical Chemistry, NewFood. Demonstrations and dedicated sessions on the relevant conferences Presenting results at established international *photonics conferences* such as SciX, SPEC, Food Safety, PITTCON, SPIE Photonics West, SPIE Photonics Europe, ANALYTICA, PITTCON; *food science and technology conferences* - Hungalimentaria 2021 (Budapest), Conference on Food Science and Technology 2020 (Budapest, Szent István University), Food, (RAFA Prague) Feed (World Nutrition Forum); *analytical conferences*, such as the World Mycotoxin Forum, EuroAnalysis2021 (www.euroanalysis2021.nl).	Scientific community | NMBU	Wide dissemination of PHOTONFOOD results, peer scrutiny and validation of results |10 publications, 20 conference presentations
Project cluster event – A project cluster and networking event held together with similar projects of the ICT-37-2020 call after the first project year to present initial results from the project to coordinate research developments and facilitate the efficient use of standard approaches to technology development and food safety.	Scientific community, researchers and EU projects | accelCH	Exchange of knowledge with other relevant national and EU projects and research groups
Training activity - Partners will organise trainings and training materials derived from results and outcomes from PHOTONFOOD to students through existing training events and programs. These will include the following: **NEBIH** takes part in university education with two external departments jointly operated with University of Veterinary Medicine, Budapest and Szent István University, Faculty of Food Science. This will include in Bachelor/Master courses, Lab settings for students/researchers, online webinars for food industry. **NP** will provide training in high-tech photonics within summer course "Bachelor of Nanotechnology" from the University of Würzburg. NP also will present on summer schools for students e.g. the iNOW (International Nano-Optoelectronics Workshop) summer; Winter School Mautendorf "New developments in solid state physics". **WU** as a co-organizer of the world's leading symposium series Recent Advances in Food Analysis (RAFA) will organize European Workshop on Portable Food Analysis and Citizen Science in the scientific program on RAFA2021 and RAFA2023. Moreover, WU will have a separate section for live demos in the RAFA Smart Lab demo sessions at RAFA2021/2023 (see www.rafa2021.eu and to get an idea about how it will look like, check out www.rafa2019.eu). **BOKU** have two courses to which it would be added a PHOTONFOOD workshop and/or lecture, i.e. "Biohazards in Food" within our Master Programme "Safety in the Food Chain" and to 2 weeks training course "Mycotoxin Summer Academy".	Students, researchers | NEBIH	Knowledge distribution to the next generation of food safety experts | each listed partner organised min 1 training event.

Policy brief - Deliver concise policy briefs to support national and international standard setting and policy processes Policy briefs will be made available via the project website, partner and relevant authority platforms and presented at policy related events and standards working groups.	Regulators & policy makers, relevant authorities | NEBIH	Possible influence and contributions to standards and policy | guidelines distributed/ downloaded 80 times
White paper - Develop a White Paper on 'photonics-based novel technologies for food safety applications'' in the final year. Provide recommendations to be considered for future guidelines and international regulations Policymakers will be invited to comment on the draft paper, which will be discussed before a final decision is made and presented during the EU Parliament workshop	Regulators & policy makers, relevant authorities | NEBIH	Inform food safety professionals about new applications and guidelines; brief policy makers; contribute to standards
PHOTONFOOD live demonstrators - The PHOTONFOOD demonstrators and technologies developed in WP6 will be presented across Europe in form of live demos at different tradeshows and local events. NP will be demonstrating PHOTONFOOD technology on trade shows for lasers or demonstrators at Photonics West- San Francisco, Analytica- Munic, Laser World of Photonics- Munich, Sensors- Nuernberg, Pittcon, New Orleans, Mirsens workshop http://mirsensconference.org/. **UULM** will organise live demonstration sessions of the devices locally in the lab as "Industry Information Days" and also hands-on 1-2 day workshops at UULM for industry with theoretical sessions and hands-on practical sessions.	Food industry and technology providers | NP	Disseminate results, raise interest and understanding of our approach, build contacts for future uptake of results 3 demonstrator sessions
Final project event - Present final project results from PHOTONFOOD in the final 6 months of the project at a workshop in Brussels for stakeholders from DG-Reseach, DG-Sante, DG-Agri and that event will be open for members of parliament and key stakeholder group representatives from the food value chain. The workshop will consist of presentations of the key outcomes from the project, introduction to our policy recommendations and whitepaper as well as a round table session with key experts.	Policy and decision makers, EU parliament members and stakeholders from the food value chain | WU	Knowledge transfer, uptake of results and networking | 80 participants at event

Publications and Open Access Policy. In line with the Horizon 2020 rules and current trends in IP management and publication strategies, the PHOTONFOOD consortium is committed to open access publishing, following the Guidelines on Open Access to Scientific Publications and Research Data in Horizon 2020. This means all beneficiaries agree with the principle that an article is immediately provided in open access mode by the scientific publisher. As the associated costs are usually shifted away from readers, and instead to the university or research institute to which the researcher is affiliated, such costs have been accounted for by our research partner institutions. Budget has been allocated accordingly for open access fees (see section 3.4). In accordance with the rules of the Grant Agreement, partners are required to give prior notice of any planned publication.

- **Open Data Policy.** Our PHOTONFOOD consortium has agreed on a joint Open Data Policy. This implies that we will develop a **Data Management Plan** (**DMP**, D8.7, M6). The DMP will determine and explain what data, generated by our research, will be made available for open access, based on the Guidelines on Data Management in Horizon 2020, and define the rules for future use of the data. A non-exhaustive preliminary overview of the types of data we expect to generate and collect during the project has been documented in Table 10. The Data Manager will be responsible for the design and implementation of a solid plan for the collection, storage, evaluation and exploitation of the data that will be produced during and after PHOTONFOOD. Openness, accessibility to beneficiaries and reusability of the produced data will also be ensured by the data management plan. The data management plan includes all the necessary

steps for efficient collection, safe storage, deep exploitation and evaluation of the data. These steps involve the Data Repository for sharing data and the Design and Implementation of a process, which will integrate data coming from different sources (Data Mapping) and securely store them. The Data Manager will also take action for metadata handling and sensitive data handling. Furthermore, data produced throughout PHOTONFOOD will be shared among open European Research Databases. The project will comply with the newly imposed rules concerning data quality, sharing and security (also GDPR principles will be applied). The cloud-based database will be used as a first basis for exploitation of results and further exchange of knowledge and expertise. A data management plan will be produced by Month 6 and updated at the project midterm and end. We will ensure that:

- Research data will be easily discoverable, accessible, assessable and intelligible, usable beyond the original purpose for which they were collected, interoperable to specific quality standards, whenever relevant.
- All data generated in the course of this Research and Innovation Action, will be assessed for quality and quantity.
- All data will be evaluated regarding new IP that can either be protect published or used by the project partners, as described in the IP management section above.
- The Pilot on Open Research Data of Horizon 2020, which aims at improving and maximising access to and re-using of research data generated by PHOTONFOOD, is followed.

The PHOTONFOOD coordinator as WP8 lead will manage all data related activities and issues to ensure that the defined data management processes, as defined in Section 3.2, are followed by all partners.

Table 10: PHOTONFOOD preliminary types of data and their use

WP	Type of Data	Methodology
WP1	User requirements and experience data.	Interviews and questionnaires will be anonimised, processed and published in scientific report.
WP2	Scientific research and development data.	Paper microfluidics, 3D printing, surface chemistry.
WP3	Tuning Characteristics of the laser; Optical properties of the LEDs; Light-Current-Voltage Characteristics.	FTIR measurements, Standard electro-optical characterization (photodiodes, current and- voltage sources-/meters) followed by a dedicated algorithm for data analysis in case of the tunable lasers.
WP4	Pre-treatment models, calibration data. Large amounts of HI-FI and MI-FI data collected during the project. Data on food safety and food waste production.	Multivariate model development. Big data analysis.
WP5	MI-FI and HI-FI test data.	MI-FI and HI-FI device test during development of the devices.
WP6	LC-MS/MS data for reference analyses of mycotoxins in peanuts and wheat. MI-FI and HI-FI data.	Liquid Chromatography – Tandem Mass Spectrometry (LC-MS/MS) for reference analysis.

IPR management. The management of intellectual property rights (IPR) will be pursued vigorously over each step of the knowledge acquisition process with exploitation being agreed internally between the respective PHOTONFOOD partners in accordance with the management procedures as described in Section 3.2. We have agreed on the following organisation of related tasks:

- The Coordinator will monitor all developments in order to ensure that such activities are compliant with the rules of the Consortium Agreement.
- All partner will disclose all foreground IP generated in the project to the Management Board.
- The Innovation Manager (Prof. Joachim Scholderer, NMBU) will monitor the IP landscape and FTO issues, liaise with the partner TTOs and legal representatives, develop a coherent IP strategy for the project, and assess the right forms of protection for the foreground IP generated in the project.
- The Exploitation Manager (Dr. Julian Haas, HS) will closely collaborate with the Innovation Manager, ensure that the IP strategy supports the further commercialisation of the results.

In order to fully exploit the project results and to pave the way for future commercialisation, we have already agreed on the following principles that will govern the IP terms in this project, and form the basis of the IP terms in the Consortium Agreement:

- **Access rights**: In order to ensure access rights to and use of existing knowledge the consortium members will differentiate between Background and Foreground Knowledge:
- **Background**: The Beneficiaries will bring background knowledge into the project (e.g. the technologies to be used). The availability of background will be declared, and a list of available background knowledge will be maintained in the Consortium Agreement. Each Beneficiary will grant the other partner(s) free access to this knowledge for the purposes of this project and for the duration of that project.
- **Foreground**: Existing technologies will be further developed and new knowledge and products will be created during the PHOTONFOOD project. Each new result will be notified to the Management Board for dissemination and exploitation.
- **Solely generated IP:** Will be solely owned by the generating party, with an option to the other project partners of a non-exclusive license to foreground IP for commercialisation based on fair and reasonable conditions. **Jointly generated IP:** All intellectual property generated jointly during the course of the Project ("Joint Foreground IP") will be jointly owned by the generating parties and will be managed in accordance with the terms of a Joint Ownership and Management Agreement (JOMA). The terms and conditions will be fair and reasonable and follow the commercialisation strategy below.

Expected exploitable results and exploitation strategy. Although there are no existing patents or patent applications to limit the objectives or exploitation activities planned for the results of this project at present, the consortium will follow this closely through the project period.

The commercial actors in the consortium will work closely with the Innovation Manager and the Exploitation Manager to manage IP in a way that contributes to the short-term and long-term objectives of the project, and at the same time balances the different interest within the consortium with regard to commercialisation. One important aspect of this is freedom to operate: the IPR landscape will be mapped in the beginning of the project and continuously monitored throughout the project's lifetime. This will ensure that relevant background IP is duly integrated into the consortium agreement and that the necessary licenses are negotiated by the project consortium. A second important aspect is coherent IP strategy: it will commit the consortium partners to an agreed IP strategy that balances the interests of all participants (secrecy clauses, patenting, trademarking, licensing, publishing), can leverage the further commercialisation and business development activities planned after the end of the project's lifetime, and avoid undue conflicts between the consortium partners and with third parties.

Detailed report to include the consortium's plan for commercialisation of the developed technology. Develop draft exploitation agreement to build on Consortium agreement. Actively seek licensees and future funding to ensure ease of commercialisation and funding during first year of exploitation. Subcontract services of company to carry out future funding and investment searches. Begin process of establishing the product in product range, draw up distribution agreement and supply chain agreements with partners.

Commercial viability of the PHOTONFOOD technologies and innovations will be ensured by an evaluation of costs and benefits planned in all technical WPs. The following paragraphs provide a qualitative overview of the commercial interests of each PHOTONFOOD partners:

NP: NP is the world leader for single mode light sources for tunable diode laser spectroscopy (TDLAS). All research activities of NP follow the strategy to maintain this position and to open up new markets such as in the food industry and the medical sector. Traditionally, the TDLAS market is mainly focusing on industrial process control and safety applications in the oil and gas industry. In recent years, NP have observed a strong market demand for both, highly specialized widely tunable single mode laser sources for high-end sensors as well as a strong customer interest in low and medium cost high power MIR light sources for consumer or semi-professional applications. Both topics are addressed by NP within the research undertaken within the PHOTONFOOD Project. The tunable cascaded laser shall address the high-end market whereas the MIR IC-LED will serve the semiprofessional market. If the sales number grow well, the MIR IC-LED device is also expected to be cost competitive for the consumer market.

IRIS: In PHOTONFOOD, IRIS will provide a supercomputing cloud-enabled platform to allow the digitalisation from the MI-FI and HI-FI analysers and to commercialize it through a Software-as-a-Service (SaaS) business model. This has the advantage of offering affordable, high performance computing to end-users for high processing power and speeds in novel sensing technologies. During the development of the project, IRIS will structure the value proposition of the developed software. In addition, IRIS will draw up the

product datasheets that will outline the key information, and will also create software brochures outlining applications, highlighting the benefits, etc. The services that support the software will also be clearly defined (maintenance service, customization services, upgrades to the software and models), consultancy based on system use and data understanding. The software interface will also be a further vehicle to build the PHOTONFOOD brand identity: this aspect will be developed in collaboration with the other industrial partners.

IBM: In PHOTONFOOD IBM will provide an advisory help to develop a strategy for implementing the MI-FI monitoring data and HI-FI reference analysis data into the Food Trust™ block chain for big data analysis to improve food safety and reduce waste in the whole food production chain.

ROMER: ROMER is in close contact with its customers and is thus very familiar with their needs. Based on this expertise, ROMER will evaluate the HI-FI device using real world samples (in cooperation with BOKU) and the reference data available for these samples. Based on the end users requirements well known to ROMER, they will be able to provide valuable feedback on the strengths and weaknesses of the developed device and suggest improvements.

Seeberger: The PHOTONFOOD technology is expected to save waste and is economically of high interest for Seeberger. The technology which can be used for monitoring on farms is expected to reveal contamination at the origin. This would allow to avoid that contaminated raw goods would not even be sent and not mixed with non-contaminated batches. If contaminated raw goods are sent to the distributer, the entire final batch of around 20 tons is contaminated and must be blocked, destroyed or returned.

Hahn-Schickard (HS): In parallel to NP, which is taking care of the commercialization of the light sources, Hahn-Schickard will manage the commercialization of the final HI-FI and MI-FI products at the end of the project. Market demands will be evaluated critically and potential stakeholder demands will be collected in order to adapt potential products after the project to that need and with the highest value and user-friendliness for final customers.

accelCH: accelCH does not plan to directly commercialise the results of the project. They rather anticipate to increase their visibility by participating in excellent projects like PHOTONFOOD

BIGH: For the aquaponics farm BIGH, the PHOTONFOOD technologies will allow to improve their production of aromatic herbs both in terms of quantity and quality. Indeed, fungi causing Downy mildew in herbs and tomatoes is one of the main contaminants for their production. Having an early detection of this fungi could be used for taking appropriate early actions and preventive measures.

BAMA: BAMA is the Norwegian market leader within fresh fruit and vegetables. Mycotoxins pose a huge risk for BAMA. Having a monitoring and analysis tool for mycotoxin at hand would enable them to improve food safety and to proof that their products are safe.

Several results are expected to be exploitable for non-commercial purposes. The PHOTONFOOD partners have identified a number of ways in which they wish to use their research results:

- **Subsequent projects -** By using the results in further research activities, such as (i) mycotoxin monitoring, recommended for industrial partners; (ii) on-site quick testing by authority personnel; (iii) research activities, such as Horizon Europe and national funded research and development programs aiming for on-site testing of food quality and safety parameters; (iv) in Horizon Europe- and national funded research and development programs aiming for citizen science;
- **Positioning -** As a recognized expert center for official controls via on-site testing: National Reference Laboratory (NRL) and European Reference Laboratory (EURL) for on-site testing of food quality and safety parameters.
- **Publishing data** – academic publishing in peer-reviewed and popular science journals,
- **Networking – BIGH** will participate in networking with the food science hub; **NP** will participate with presentations on summer schools for students e.g. the iNOW (international nano-optoelectronic workshop) summer; Winter School Mautendorf "New developments in solid state physics"; **WU** will perform proactively sharing the PHOTONFOOD developments and outcomes through the official EU network of national and European Reference Laboratories (note that Wageningen Research is the designated EURL for mycotoxins, plant toxins and veterinary drug residues). **BOKU** will link PHOTONFOOD to the networks of the Austrian Society of Analytical Chemistry and of the International Society of Mycoxicolgy and Global Food Security networks at Queen´s U Belfast.

Exploitation measures. To facilitate the future commercial and non-commercial uptake of results based on our strategy above we have defined the following activities to exploit results during and after the project ends:

Table 11: PHOTONFOOD exploitation measures

Activity description	Target group | Lead	Impact | KPIs
Business Model– To facilitate the future upscaling and uptake of the PHOTONFOOD platform and solutions, potential business cases will be developed. This will include a stakeholder mapping as well as an extensive SWOT and PESTEL analysis of our new technologies to ensure its sustainability in the long term.	Industry, technology providers | HS	Uptake of project results after the project end | business models developed and implemented
Strategic grant planning - To support the project and all partners in the use of their results and outcomes in further research, accelCH will conduct a strategic grant plan to identify future funding opportunities for the PHOTONFOOD platform and the individual technologies developed. Early identification of potential funding opportunities will support the preparation of grant applications to exploit project results.	Researchers, project partners | accelCH	Exploitation of outcomes through follow-up projects | one planning session with all partners to identify future funding opportunities
Standardisation and guidelines: The project results have the potential to be integrated into national food safety related guidelines such as Good Hygiene Practices based on the Regulation (EC) No 852/2004 and national level food codices, such as the Codex Alimentarius Austriacus and the Codex Alimentarius Hungaricus. We will share our policy brief with decision and policy makers which will include recommendations for food safety guidelines. Our recommendations are not legally binding, but could potentially influence regulations if our results are strong which could require countries to change their laws.	Policy-decision makers | NEBIH	Use and contributions to standards and guidelines | uptake of results
Policy change - The project partners are committed to exploiting their results to have a direct influence on current guidelines on food safety. Using dissemination materials created during the project we use our established contacts with policy makers to change policies.	Policy-decision makers | NEBIH	Stakeholders briefed on our approach to contribute to change | contacts with 50 policy makers
Roadmap for exploitation (RfE): Develop a path-to-market strategy including an exploitation roadmap building upon the initial business models (see table 7)	Industry, technology providers, researchers, HS	Uptake of project results after the project end | RfE developed (D7.6, M48)

Business case. The Innovation Manager of the project has developed an initial business case in close collaboration with the other PHOTONFOOD partners. The business model canvas is shown in Figure 13.

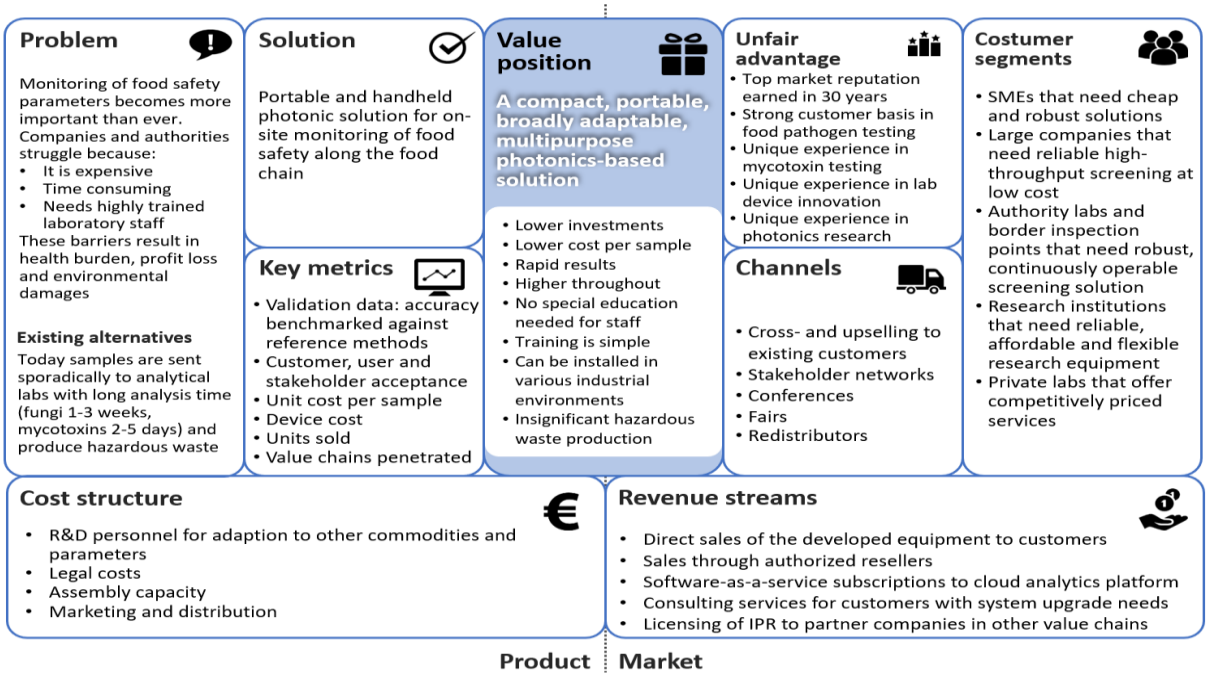

Figure 13: PHOTONFOOD business canvas

Roadmap and outlook. To achieve the expected impact of PHOTONFOOD results beyond the end of the project, we have designed a preliminary roadmap for exploitation showing the timeline and the main measures needed to facilitate the up-take of commercial and non-commercial exploitation of results. The initial roadmap for exploitation (RfE) has been visualised in Figure 14 and covers PHOTONFOOD exploitation measures during the lifetime of the project (short-term) and mid- and long-term plans (<4 years and <10 years after the project ends).

The initial RfE will be used as a basis for the achievement and wide uptake of PHOTONFOOD project results and outcomes and further developed during the project based on the research results from WP1-WP6 and the business plans. The results will be recorded in the deliverable Roadmap for Exploitation (D7.6, M48) and used by all partners to facilitate the implementation of exploitation measures beyond the duration of the project. This will include: (a) a detailed report of the consortium's plan for commercialisation of the developed technology, (b) a draft exploitation agreement to be annexed to the consortium agreement, (c) active seeking of licensees, (d) active seeking of future funding to leverage commercialisation and funding during first year of exploitation, (e) initiating the process of establishing the product/service range, and (f) initiation of up distribution agreement and supply chain agreements with partners.

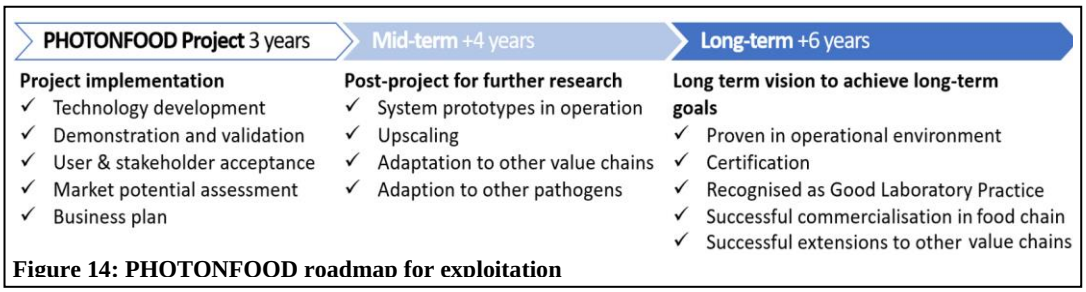

Figure 14: PHOTONFOOD roadmap for exploitation

2.2.2 Communication activities

The PHOTONFOOD communication activities target a broader audience than dissemination measures (as described in Section 2.2.1.1). They are aimed on the one hand to support and complement our defined dissemination measures, and on the other hand offer materials that can be further spread by our partners' network. Communication activities aim at: Raising awareness of the project to farmers/producers, processors, logistics, retailers and technology developers along the food value chain, as well as the general public on the

importance of early contamination detection in food and the potential of the PHOTONFOOD solution. In order to achieve these aims we have planned the communication measures presented in Table 12.

Table 12: PHOTONFOOD communication activities

Activity Description	Target group | Lead	Impact | KPIs
Project website - Set up and maintain the project website, which will notably function as the main source of information and resources related to the project, partners, latest developments, publications and public deliverables for all stakeholders. Information relevant to the project will be updated regularly including short blog texts on the project's progress as well as events our partners are contributing to. Links to social media to for sharing relevant information through partner's existing and well-established networks on Twitter, Facebook and YouTube such as BOKU's twitter which received over 100.000 views this year.	All Target groups | accelCH	Updating of stakeholders on development and results, and raising general awareness | Average 500 users per month
Communication materials - Logo, design guidelines and templates created to support the visual identity of the project. Print and online materials created and available to all project partners to be distributed online, via social media, at conferences/exhibitions such as: • Case studies to show the food along the value chain and our solution • Poster and roll-ups for events and tradeshows • Technology factsheets on the innovative developments • Factsheet and/or brochures providing basic project information	All Target groups, but with custom-made materials | accelCH	Spreading general knowledge on the project | 200 downloads or print outs distributed
Local science events - Our academic partners organise local science events to inform the general public on science related topics and increase the interest of students to a career in science. Participate at local events to facilitate two-way engagement with society and increase interest in the project in students and the general public.	General public, students, media | UULM	Increase interest in science and the project | participation at 4 events
Audio/visual material - Interactive digital audio/visual material will be created including videos presenting the project activities including interviews with key partners and infographics to visualise complex information in a simple way to the food industry the general public. The digital media will be shared via the project website, social media channels and existing partner networks. Videos and infographics will be distributed via the project website, partners' social media channels	General public and media, Food industry | accelCH	Inform dedicated target groups about the PHOTONFOOD approach | 600 views online
Press releases and articles - As interest in food waste and food safety is increasing, communications with mass media will prove central to reach out to a large audience. In case of breakthrough findings, we will issue press releases to popular media in collaboration with the media relation units of the partner institutions, will then coordinate all requests for press conferences, interviews, press, news and TV coverage	General public, media | WU	Awareness raising and informing of key project results | 4-7 press releases planned

Communication requirements. This project is funded by one of the calls under the Photonics Public Private Partnership (PPP). All communication activities related to the project will acknowledge the context of the Photonics PPP, for example by stating that the project is an initiative of the Photonics Public Private Partnership.

Specifically, for workshops, press releases, presentations etc., the EU emblem and Photonics21 logo will be displayed prominently together with the text "Photonics Public Private Partnership". The link www.photonics21.org will also be included. When communicating on Twitter or other social media about project activities, #Photonics will be included together with @Photonics21 and @PhotonicsEU.

The consortium will issue a professional press release at project launch, additional professional press releases whenever the project has reached a significant milestone or exceptional scientific, economic or societal impact is expected, and a final professional press release at project end.

A professional communication kit about the project (as a minimum a narrative text, photographs, slides possibly complemented by a video and any other suitable communication material, accompanied with copyright licences for the European Commission and for Photonics21) will be prepared and delivered *at the end of project month 3* at the latest. The communication kit will target audiences beyond the project's own

community, including the broader public and potential end-users, and focus on expected outcomes and related socio-economic benefits for the EU. The professional communication kit will be updated *at mid-term and at project end* to reflect project progress achieved.

The communication activities will involve public relations experts to ensure the pertinence of messages to target audiences and high outreach at both national and EU level through the use of the relevant national and EU-wide media channels (e.g. newspapers and broadcast).

Project communications such as project press releases, workshops announcements, websites, brochures will respect the principle of fair visibility for all partners. If the logo of any individual beneficiary is included, then the logos of all beneficiaries will be included.

3. Implementation

3.1 Work plan — Work packages, deliverables

3.1.1. Overall structure of the work plan

The PHOTONFOOD project will be implemented by an interdisciplinary consortium of academic, authority and industry partners covering 14 organizations from eight European countries. The project duration will be 48 months and the breakdown of the planned work into work packages (WPs) is based on accomplishing related work, rather than on single tasks that occur at the same time. Therefore, several WPs are executed in parallel throughout the project. The project consists of eight work packages (WPs) (see Fig. 15), where the first WP1 is related to the evaluation of the technology acceptance and techno-economical evaluation, four WPs (WP2 – WP5) focused on the photonic technology developments and developing advanced data analysis tools and software applications, WP6 will perform validation and demonstration of PHOTONFOOD technology, WP7 and WP8 are related to dissemination, communication, exploitation and management activities, respectively (Fig. 15).

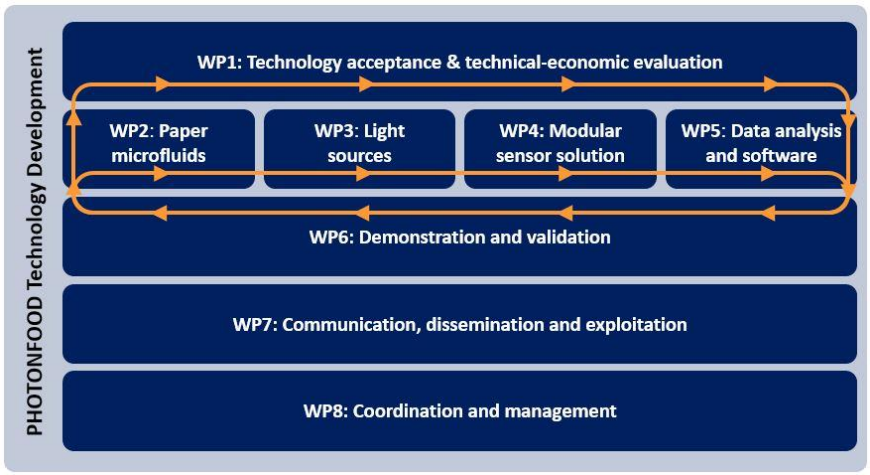

Figure 15: Overall structure of the workplan

The general overview of WPs is provided below, the list of the WPs and a detailed description of each WP are provided in Table 3.1a and 3.1b, respectively.

Assisting WPs: WP1 "Technology acceptance and techno-economical evaluation" will conduct comprehensive study on evaluating of user expectations, experience and technology acceptance of photonic-based technologies for identifying microbiological and chemical contaminations. WP1 will ensure that PHOTONFOOD technology is developed in accordance with user requirements. WP7 "Communication, dissemination and exploitation" will perform adequate dissemination and exploitation of PHOTONFOOD technology and will be responsible for the communication with the relevant audience. WP8 "Coordination and Management" will be responsible for the project coordination and management for the successful implementation of all project work activities.

Technological WPs: Four technological WPs will be responsible for the successful development of PHOTONFOOD technology. WP2 "Paper Microfluidics" focuses on the development of extraction /separation / up-concentration technology for separation of analyte markers of microbial and chemical contamination. WP2 will be strongly connected to photon-based device developing WPs, WP3 and WP4. WP3 "Novel MIR light sources" will focus on the development of cascaded gain materials forming the basis for advanced light emitting diode (IC-LEDs) and monolithically tunable laser light sources. WP4 "Modular sensor platform" will implement the results of WP3 and will perform development of portable and field-deployable (prototype) MI-FI and HI-FI analyzers. WP5 "Data analysis and software development" will focus on developing and utilizing advanced data analysis tools for the spectral-based detection and identification of microbiological and chemical contamination in different food matrices. In addition, WP5 will be focused on developing software application.